Symbolic Knowledge Discovery

FCA and pattern mining for actionable knowledge discovery

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IST, U.Lisbon, INESC-ID & IDMC, U.Lorraine, CNRS, Inria, LORIA

Based on references and notes by M. Kaytoue, S. Kuznetsov, A. Napoli, Sergei Obiedkov

Motivation & some mathematical background

Example I: A scientific question

Understanding the olfactory system

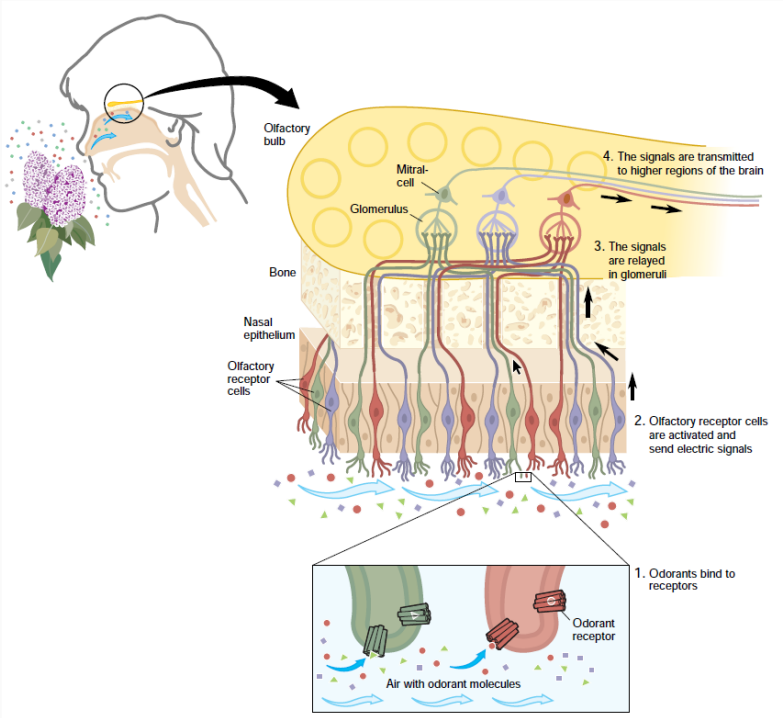
- Olfaction: ability to perceive odors
- Complex phenomenon from molecule to perception (1991)

Challenges:

- Links between physico-chemical properties and olfactory qualities (2004,2017)
- Difficulties to formulate rules

Impact:

- Fundamental neuroscience research
- Industry (food, perfume)
- Health (anosmia, ...)
- NLP (text/topic classification)



Example I: Smell, a luxury?

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L'odorat, victime du COVID-19 à l'étude

20 avril 2020 RECHERCHE

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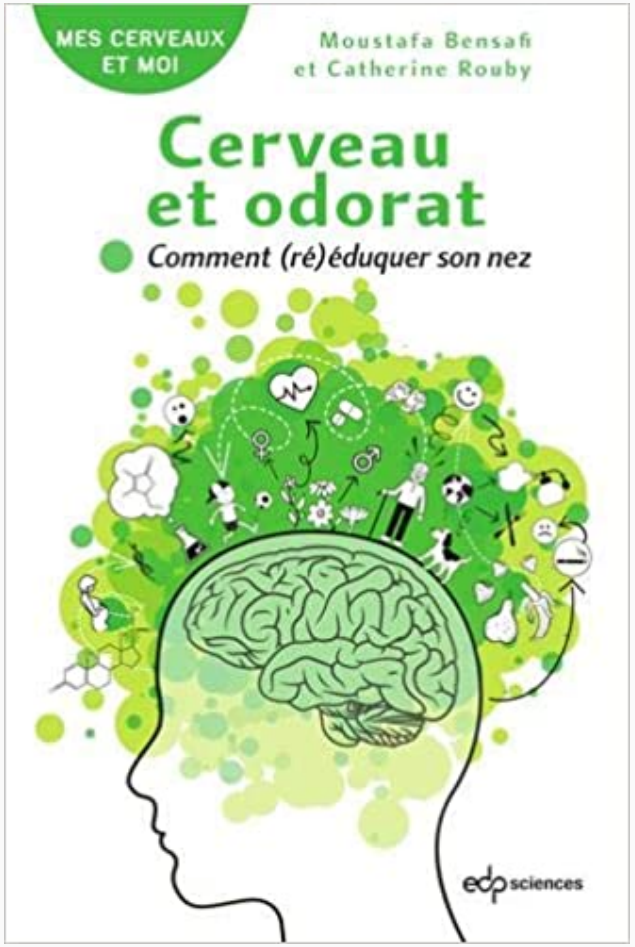
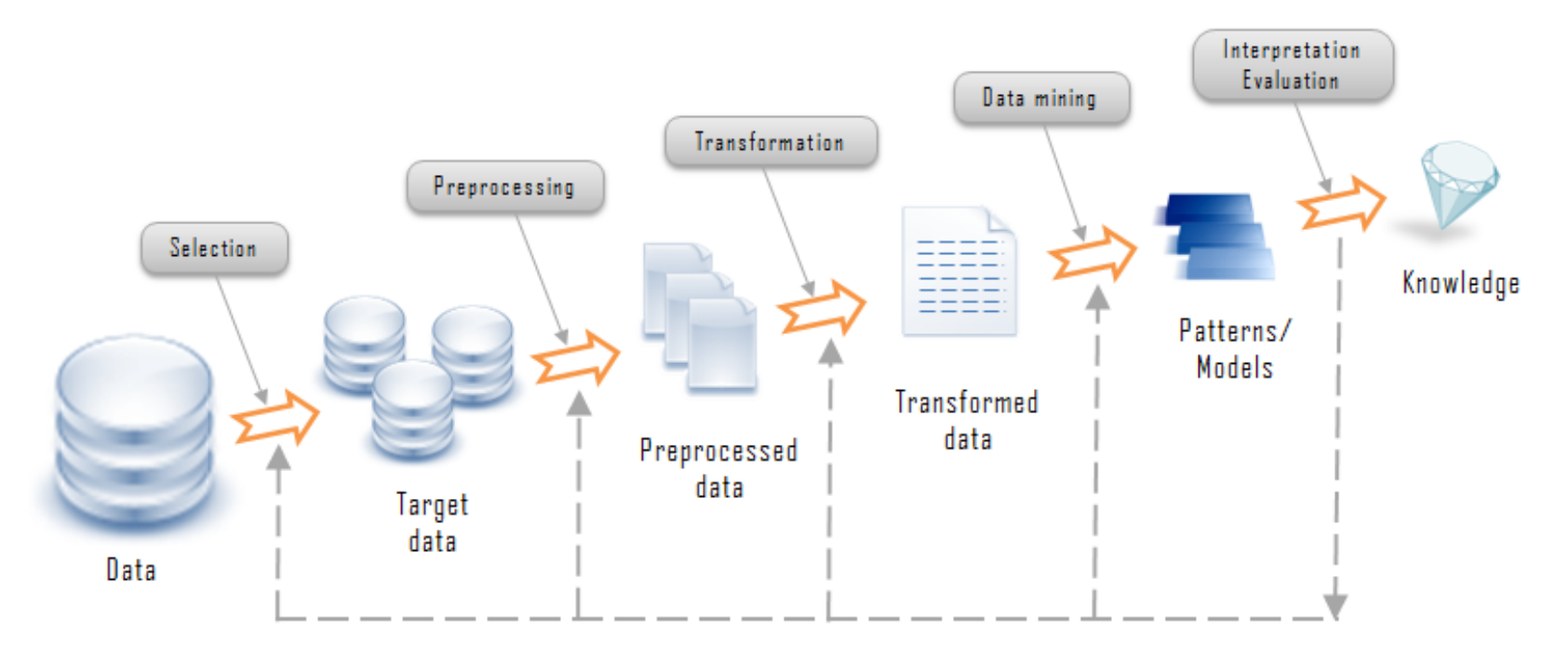
Certains patients atteints du COVID-19 ont signalé une perte de l'odorat. En partenariat avec d'autres structures françaises, le Centre de recherche en neurosciences de Lyon lance un site internet d'information et invite le grand public à participer à une étude sur ce sens méconnu. Les neuroscientifiques Moustafa Bensafi et Camille Ferdenzi, du Centre de recherche en neurosciences de Lyon, répondent à nos questions.

A - / A+

Partager ce contenu



Eliciting hypotheses from data: A KDD task



The KDD process is iterative, interactive and guided by an analyst

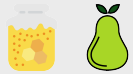
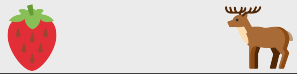
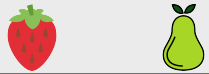
Eliciting hypotheses from data: A KDD task

Data collection

- Descriptions: chemical properties, molecular structure, ...(2005)
- Targets: odor(s), valence...(1994)

Mining (a few and good) hypotheses (in a large search space)

- Clusters, concepts, association rules, redescriptions, ...
- Mining subgroups discriminating a target attribute (1997)
  Data query: $s = nAT \geq 24 \wedge nC \leq 11$
  Query result: $support(s) = \{24, 48, 1633\}$
  $Quality(s \rightarrow pear)$ is high: all the pears, only the pears

ID	MW	nAT	nC	Odor
1	150.19	21	11	
24	128.24	29	11	
48	136.16	24	9	
60	152.16	23	11	
82	151.28	27	12	
1633	142.22	27	10	

This course...

In our investigations:

- What do we mine?
- How do we mine the best patterns?
- For what purpose?

Evaluation: Project on symbolic knowledge discovery & RAG-like tools (i.e., KG construction, ontology design)

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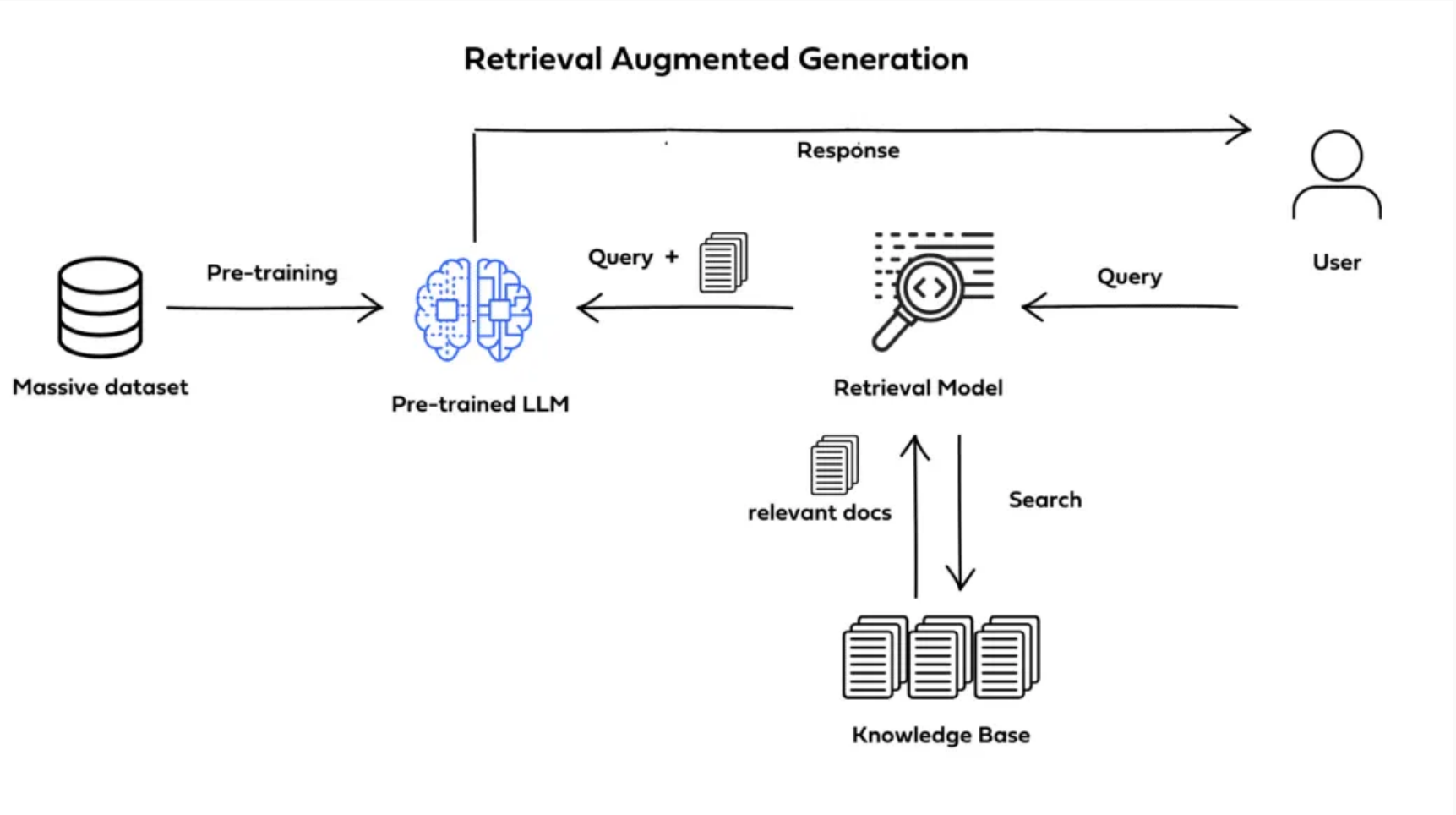
Retrieval-Augmented Generation (RAG)



From https://medium.com/@krtarunsingh/

introduction-to-retrieval-augmented-generation-rag-and-its-transformative-role-in-ai-c07e35da7f01

Case study

Objective: Enrich a domain-specific KG with symbolic data and implement a GraphRAG pipeline

1. Find a dataset of your interest (tabular data)
2. Convert your dataset into turtle format (KG)
3. Use this graph representation when querying your data with graphRAG. Comment your results.
4. Now find concepts and implications using the tools seen in the course
5. Integrate them in your KG (we call this enriched KG "ontology")
6. Use this ontology for querying your data with GraphRAG. Comment your results.
7. Propose an experimental setting (questions, metrics, etc.) to compare both approaches: KG vs. Ontologies!

Format: Work in groups (at most 3 students per group)

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  WebPage: `https://members.loria.fr/mcouceiro/`
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  Email: `egor.dudyrev@loria.fr`,
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Some mathematical background:
Ordered sets and lattices

Some mathematical background: orders and ordered sets

Let P be a set. An partial order on P is a binary relation $\leq$ on P such that:

- $x \leq x$ (reflexivity),
- $x \leq y$ and $y \leq x$ imply $x = y$ (antisymmetry),
- $x \leq y$ and $y \leq z$ imply $x \leq z$ (transitivity), for every $x, y, z \in P$.

If $\leq$ is reflexive and transitive, then it is called a quasi-order.
A linear order is a partial order $\leq$ such that, for every $x, y \in P$, $x \leq y$ or $y \leq x$.

An equivalence is a relation $\equiv$ that is reflexive, symmetric and transitive.

A set equipped with a partial order (resp. quasi-order, linear order) is said to be a partially ordered set or, simply, a poset (resp. quasi-, linearly ordered set).

NB: Every quasi-order $\leq$ induces the equivalence: $x \equiv y$ iff $x \leq y$ and $y \leq x$.

Example: For any set X, the powerset $2^X = \mathcal{P}(X)$ of all subsets of X is a poset with the partial ordering $\leq$ defined by: for $A, B \subseteq X$, $A \leq B$ iff $A \subseteq B$.

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Some mathematical background: Upper and lower bounds

Let P be an ordered set and let $S \subseteq P$. An element $x \in P$ is

- an upper bound of S if $s \leq x$, for all $s \in S$.
- a lower bound of S if $x \leq s$, for all $s \in S$.

The set of all upper bounds of S is denoted by S^u ("S upper"):

$$S^u := \{x \in P | (\forall s \in S)\, s \leq x\}$$

The set of all lower bounds of S is denoted by S^ℓ ("S lower"):

$$S^\ell := \{x \in P | (\forall s \in S)\, s \geq x\}$$

Some mathematical background: Suprema and infima

Let P be an ordered set and let $S \subseteq P$. An element $x \in P$ is called the

- least upper bound of S (or supremum) of S if for every $y \in S^u \setminus \{x\}$, $x \leq y$. For $x, y \in P$, their supremum is usually denoted by $x \vee y$ (the **join**).

  NB: if it exists, it is unique!

- More generally, the supremum of $S \subseteq P$ (when it exists) is denoted by $\bigvee S$ (join of S).

- greatest lower bound of S (or infimum) of S if for every $y \in S^\ell \setminus \{x\}$, $y \leq x$. For $x, y \in P$, their infimum is usually denoted by $x \wedge y$ (the **meet**).

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Some mathematical background: Lattices and complete lattices

A lattice is a poset L s.t. for every pair $x, y \in P$, both $x \vee y$ and $x \wedge y$ exist.

A lattice L is said to be complete if both $\bigvee S$ and $\bigwedge S$ exist, for every $S \subset L$.

Example: The powerset $2^X = \mathcal{P}(X)$ endowed with $\subseteq$ is a (complete) lattice.

Lattices as algebras: A lattice is an algebraic structure $\langle L; \vee, \wedge \rangle$ such that

- the binary operations $\vee$ and $\wedge$ are commutative and associative, and
- for every $x, y \in L$, $x \wedge (x \vee y) = x = x \vee (x \wedge y)$.

Exercise. Let $\langle L; \vee, \wedge \rangle$ be a lattice. Show that both $\vee$ and $\wedge$ are idempotent, and that the relations defined by

- $x \leq_1 y$ iff $x \vee y = y$
- $x \leq_2 y$ iff $x \wedge y = x$

are partial orders. Are they the same?

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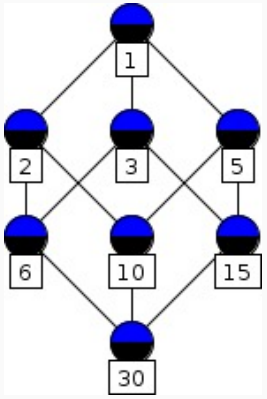
Further examples of lattices

$\mathbb{N}$ with the divisibility relation is a lattice:

- $x \leq y$ iff y is a divisor of x as integers (Euclidean division).
- $x \vee y = gcd(x, y)$ (greatest common divisor)
- $x \wedge y = lcm(x, y)$ (least common multiple)

The lattice of the divisors of 30.



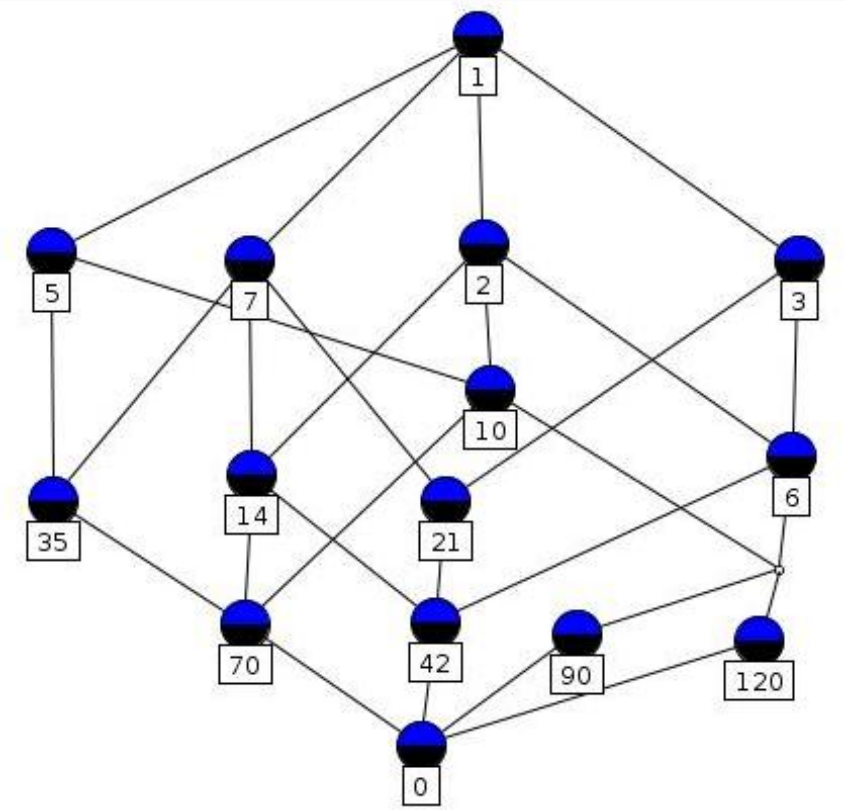
An example of classification-based reasoning

The classification process of a concept C or an individual $\mathbf{x}$ in a concept hierarchy relies on three main steps:

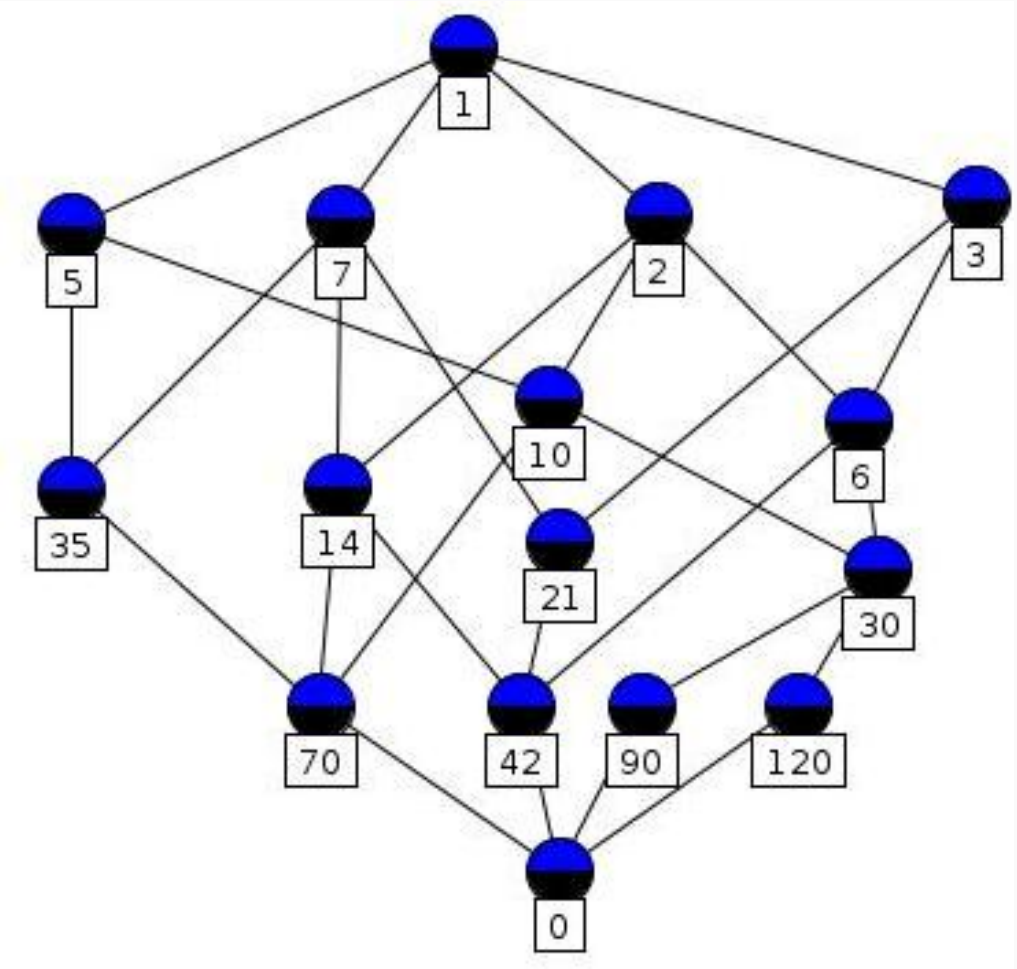
- search for the most specific subsumers (MSS) of $\mathbf{x}$,
- search for the most general subsumees (MGS) of $\mathbf{x}$
  (this step is not necessary for an individual),
- insertion of $\mathbf{x}$ in the concept hierarchy, under its MSS and over its MGS.

Classification in the division poset

Problem: Classify 30 in the following division poset.



Classification in the division poset



Outline

1. Motivation & some mathematical background
2. FCA & Algorithms
3. Implications & functional dependencies
4. Association rules & hypotheses
5. Recall the Formal Concept Analysis
6. Interesting descriptions
7. Pattern Structures

PART I

FCA & Algorithms

Formal Concept Analysis: motivation

Formal Concept Analysis (FCA) is a mathematical formalism based on lattice theory, classification and concept discovery.

FCA follows a human centered approach and supports exploration operations through the concept lattice:

- **Discovery of concepts**, i.e. classes of individuals with a description.
- **Organization of concepts** into a poset based on a subsumption relation.
- It supports exploration, e.g. information retrieval, visualization, ...

The FCA process

- The basic procedure of FCA is based on a simple representation of data, i.e. a binary table called a formal context.
- Each formal context is transformed into a mathematical structure called concept lattice.
- The information contained in the formal context is preserved.
- The concept lattice is the basis for data analysis.

Graphical representation: analysis, mining, visualization, interpretation. . .

Formal Contexts

Let G and M be sets. Elements of G are called **objects**, elements of M are called **attributes**.

Let $I \subseteq G \times M$ be a binary relation. If $(g, m) \in I$, means that object g has attribute m. The triple $\mathbb{K} \stackrel{\text{def}}{=} (G, M, I)$ is called a **(formal) context**.

Example: 2D figures

	G \ M	a	b	c	d
1		×			×
2		×		×	
3			×	×	
4			×	×	×

Galois connections and derivations

Given $\mathbb{K} := (G, M, I)$ consider two mappings $\varphi\colon 2^G \to 2^M$ and $\psi\colon 2^M \to 2^G$:

$$\begin{aligned} \varphi(A) &\stackrel{\text{def}}{=} \{m \in M \mid gIm \ \text{ for all } g \in A\}, \\ \psi(B) &\stackrel{\text{def}}{=} \{g \in G \mid gIm \ \text{ for all } m \in B\}. \end{aligned}$$

These mappings are usually called derivations.

Properties: for all $A_1, A_2 \subseteq G$, $B_1, B_2 \subseteq M$:

1. $A_1 \subseteq A_2 \Rightarrow \varphi(A_2) \subseteq \varphi(A_1)$ (order-reversing)
2. $B_1 \subseteq B_2 \Rightarrow \psi(B_2) \subseteq \psi(B_1)$ (order-reversing)
3. $A_1 \subseteq \psi(\varphi(A_1))$ and $B_1 \subseteq \varphi(\psi(B_1))$ (extensivity)

Thus: φ and ψ define a **Galois connection** between $(2^G, \subseteq)$ and $(2^M, \subseteq)$, and $A \subseteq \psi(B) \Leftrightarrow B \subseteq \varphi(A)$

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Properties of derivations $(\cdot)'$

In FCA: we unify φ and ψ into a single notation $(\cdot)'$

$$\begin{aligned} A' &\stackrel{\text{def}}{=} \{m \in M \mid gIm \ \text{ for all } g \in A\}, \\ B' &\stackrel{\text{def}}{=} \{g \in G \mid gIm \ \text{ for all } m \in B\}. \end{aligned}$$

Properties: For $\mathbb{K} = (G, M, I)$, $A, A_1, A_2 \subseteq G$, we have

- $A \subseteq A''$
- $A''' = A'$
- $(A_1 \cup A_2)' = A_1' \cap A_2'$;

Similar properties hold for subsets of attributes.

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A first concrete and culinary example

Pizzas\Toppings	tomato	onion	oregano	mozzarella	palm heart
filadelfia		x	x		x
regina	x		x	x	
roma	x	x	x	x	
margherita	x			x	
napoli	x	x	x	x	
castelli	x		x	x	

Computing the images of sets of objects and attributes

{g2}' = {m1, m3, m4}:

G \ M	m1	m2	m3	m4	m5
g1		x	x		x
g2	**x**		**x**	**x**	
g3	x	x	x	x	
g4	x			x	
g5	x	x	x	x	
g6	x		x	x	

Computing the images of sets of objects and attributes

{m3}' = {g1, g2, g3, g5, g6}:

G \ M	m1	m2	m3	m4	m5
g1		x	**x**		x
g2	x		**x**	x	
g3	x	x	**x**	x	
g4	x			x	
g5	x	x	**x**	x	
g6	x		**x**	x	

Computing the images of sets of objects and attributes

{g3,g5}' = {m1, m2, m3, m4}:

G \ M	m1	m2	m3	m4	m5
g1		x	x		x
g2	x		x	x	
g3	**x**	**x**	**x**	**x**	
g4	x			x	
g5	**x**	**x**	**x**	**x**	
g6	x		x	x	

Computing the images of sets of objects and attributes

{m3,m4}' = {g2, g3, g5, g6}:

G \ M	m1	m2	m3	m4	m5
g1		x	x		x
g2	x		**x**	**x**	
g3	x	x	**x**	**x**	
g4	x			x	
g5	x	x	**x**	**x**	
g6	x		**x**	**x**	

Formal concepts

(Formal) concept is a pair (A, B) such that $A \subseteq G,\ B \subseteq M$ **and**

$$A' = B \quad \text{and} \quad B' = A$$

A is called the extent **and** B the intent of the concept (A, B).

Concepts are partially ordered by the relation

$$(A_1, B_1) \geq (A_2, B_2) \iff A_1 \supseteq A_2 \quad (B_2 \supseteq B_1).$$

This partially ordered set constitutes the concept lattice $\underline{\mathfrak{B}}(G, M, I)$

NB: Formal concepts correspond to maximal rectangles in contexts

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Back to our example...

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1		×			×
2		×		×	
3			×	×	
4			×	×	×

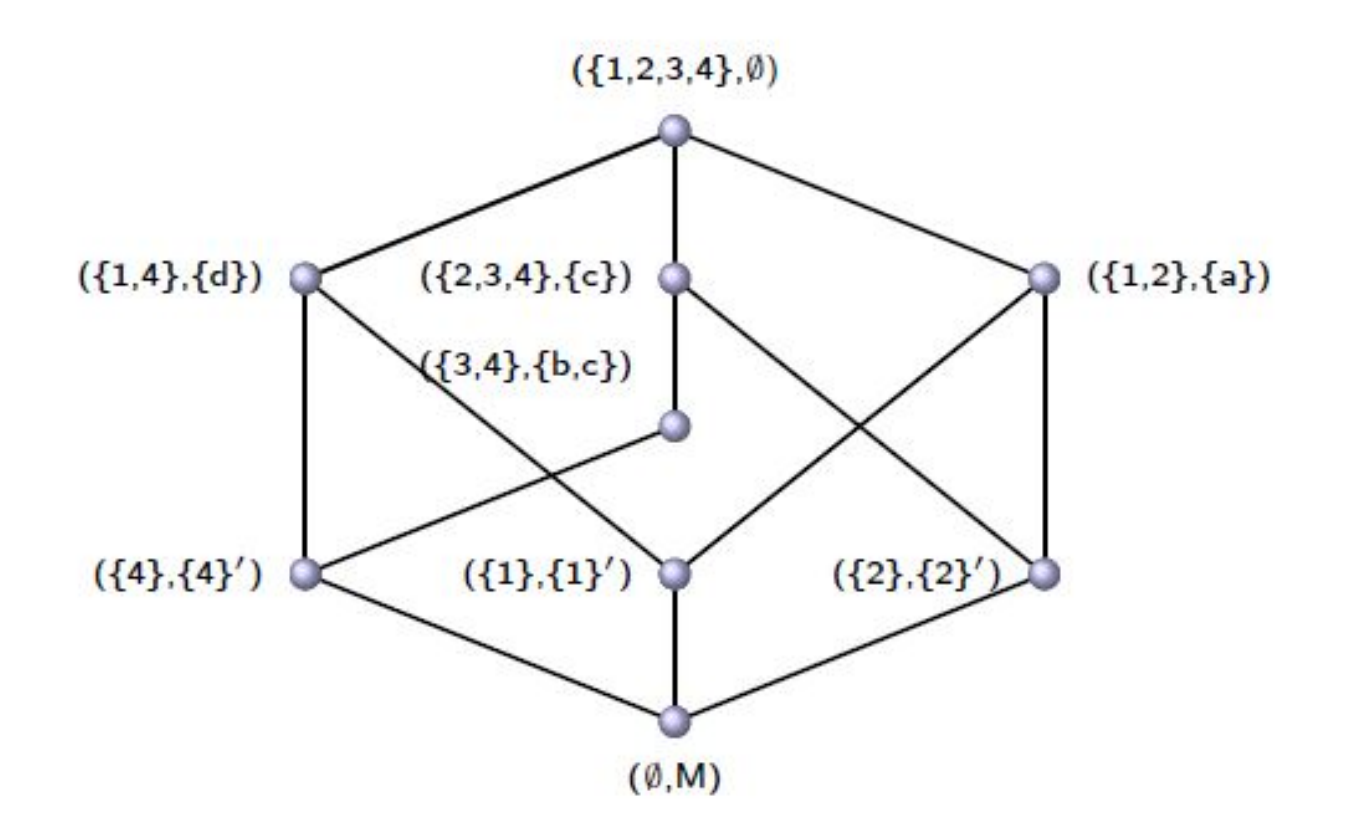
Objects:

- **1** – equilateral triangle
- **2** – rectangle triangle
- **3** – rectangle
- **4** – square

Attributes:

- **a** – has 3 vertices
- **b** – has 4 vertices
- **c** – has a right angle
- **d** – equilateral

The corresponding concept lattice



	G \ M	a	b	c	d
1		×			×
2		×		×	
3			×	×	
4			×	×	×

a – has 3 vertices

b – has 4 vertices

c – has a direct angle

d – equilateral

Closure systems...

Concept lattices are a noteworthy data science example of a closure system:

- a set of descriptions of some objects from a domain
- together with all possible intersections of such descriptions.

These intersections give a similarity of objects in terms of descriptions.

Closure operator on a set

A closure operator on set G is a mapping $\varphi\colon \mathcal{P}(G) \to \mathcal{P}(G)$ such that:

1. $\varphi(\varphi(X)) = \varphi(X)$ (idempotency)
2. $X \subseteq \varphi(X)$ (extensity)
3. $X \subseteq Y \Rightarrow \varphi(X) \subseteq \varphi(Y)$ (monotonicity)

For a closure operator φ the set φX is called closure of X.

A subset $X \subseteq G$ is called closed if $\varphi X = X$.

NB: Both $(\cdot)'' \colon 2^G \to 2^G$ and $(\cdot)'' \colon 2^M \to 2^M$ are closure operators.

The study of such closure systems is the main focus of FCA.

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The study of such closure systems is the main focus of FCA.

The basic theorem of FCA

$\underline{\mathfrak{B}}(G, M, I)$ is a complete lattice where the join and the meet are given by:

- $\bigvee_{k \in K}(A_k, B_k) = ((\bigcup_{k \in K} A_k)'', \bigcap_{k \in K} B_k)$
- $\bigwedge_{k \in K}(A_k, B_k) = (\bigcap_{k \in K} A_k, (\bigcup_{k \in K} B_k)'')$

NB: the intersection of closed sets is a closed set but the union is not necessarily a closed set.

A simple algorithm to construct concept lattices:

The algorithm of Bordat

(not covered)

From contexts to concept lattices

Let us consider a set of objects G and a set of attributes M.

- The Bordat algorithm is able to build the formal concepts and the order between the concepts, thus the whole concept lattice.
- Let us consider a set of objects G and a set of attributes M.
- Set the bottom concept as the pair $\perp = (M', M)$ unless there is one object owning all attributes (then $M' \neq \emptyset$).

The Bordat algorithm

Let $C_k = (X_k, Y_k)$ be a formal concept.

The concept $C_{k+1} = (X_{k+1}, Y_{k+1})$ is a subsumer of C_k in the lattice whenever:

(i) $X_{k+1} = X_k \cup \{x \in G \setminus X_k \ : \ f_{C_k}(x) = Y_{k+1}\}$ where f_{C_k} computes the intent of x restricted to Y_k

NB: extensions are growing, while intensions are shrinking!

(ii) $Y_{k+1} \in Max\{f_{C_k}(x) \ : \ x \notin X_k\}$

The Bordat algorithm

G \ M	a	b	c	d	e
g1		x	x		x
g2	x		x	x	
g3	x	x	x	x	
g4	x			x	
g5	x	x	x	x	
g6	x		x	x	

- $G = \{g1, g2, g3, g4, g5, g6\}$ and $M = \{a, b, c, d, e\}$
- $\perp = (\emptyset, M) = (X_0, Y_0)$

The Bordat algorithm

- $G = \{g1, g2, g3, g4, g5, g6\}$ and $M = \{a, b, c, d, e\}$
- $\perp = (\emptyset, M) = (X_0, Y_0)$

Finding the subsumers of $(X_0, Y_0) = (\emptyset, M)$:

- $X_1 = X_0 \cup \{x \in G \setminus X_0 \ : \ f_{C_0}(x) = Y_1\}$
- $Y_1 \in Max\{f_{C_0}(x) \ : \ x \notin X_0\}$
- $f_{C_0} \subseteq (G \setminus \emptyset) \times \{a, b, c, d, e\}$
- $\{f_{C_0}(x) \ : \ x \notin X_0\} = \{f_{C_0}(x) \ : \ x \in G\} =$
  $\{f_{C_0}(g1), f_{C_0}(g2), f_{C_0}(g3), f_{C_0}(g4), f_{C_0}(g5), f_{C_0}(g6)\} =$
  $\{\{b, c, e\}, \{a, c, d\}, \{a, b, c, d\}, \{a, d\}, \{a, b, c, d\}, \{a, c, d\}\}$

Possible intents Y_1:

- $Y_1 \in Max\{f_{C_0}(x) \ : \ x \notin X_0\} = \{\{b, c, e\}, \{a, b, c, d\}\}$

The Bordat algorithm

- $Y_1 = \{b, c, e\}$
  $X_1 = X_0 \cup \{x \in G \setminus X_0 \ : \ f_{C_0}(x) = Y_1\} =$
  $\{x \in G \ : \ f_{C_0}(x) = \{b, c, e\}\} = \{g1\}$
- $Y_2 = \{a, b, c, d\}$
  $X_2 = X_0 \cup \{x \in G \setminus X_0 \ : \ f_{C_0}(x) = Y_2\} =$
  $\{x \in G \ : \ f_{C_0}(x) = \{a, b, c, d\}\} = \{g3, g5\}$

Direct subsumers of $C_0 = (X_0, Y_0)$:

- $C_1 = (X_1, Y_1) = (\{g1\}, \{b, c, e\})$ and
- $C_2 = (X_2, Y_2) = (\{g3, g5\}, \{a, b, c, d\})$

Direct subsumers of $C_1 = (X_1, Y_1) = (\{g1\}, \{b, c, e\})$:

- $X_3 = X_1 \cup \{x \in G \setminus X_1 \ : \ f_{C_1}(x) = Y_3\}$
- $f_{C_1} : (G \setminus \{g1\}) \times \{b, c, e\}$
- $Y_3 \in Max\{f_{C_1}(x) \ : \ x \in \{g2, g3, g4, g5, g6\}\}$
- $Y_3 \in Max\{f_{C_1}(g2), f_{C_1}(g3) f_{C_1}(g4), f_{C_1}(g5), f_{C_1}(g6)\} =$ $Max\{\{c\}, \{b, c\}, \emptyset, \{b, c\}, \{c\}\} = \{b, c\}$

Hence:

- $C_3 = (X_3, Y_3) = (\{g1, g3, g5\}, \{b, c\})$

The Bordat algorithm

Direct subsumers of $C_2 = (X_2, Y_2) = (\{g3, g5\}, \{a, b, c, d\})$:

- $X_4 = X_2 \cup \{x \in G \setminus X_2 \ : \ f_{C_2}(x) = Y_4\}$
- $f_{C_2} : (G \setminus \{g3, g5\}) \times \{a, b, c, d\}$
- $Y_4 \in Max\{f_{C_2}(x) \ : \ x \in \{g1, g2, g4, g6\}\}$
- $Y_4 \in Max\{f_{C_2}(g1), f_{C_2}(g2), f_{C_2}(g4), f_{C_2}(g6)\} =$
  $Max\{\{b, c\}, \{a, c, d\}, \{a, d\}, \{a, c, d\}\} = \{\{b, c\}, \{a, c, d\}\}$

Hence:

- the first subsumer is identical to $C_3 = (X_3, Y_3) = (\{g1, g3, g5\}, \{b, c\})$
- the second subsumer $C_4 = (X_4, Y_4) = (\{g2, g3, g5, g6\}, \{a, c, d\})$

Direct subsumers of $C_3 = (X_3, Y_3) = (\{g1, g3, g5\}, \{b, c\})$:

- $X_5 = X_3 \cup \{x \in G \setminus X_3 \ : \ f_{C_3}(x) = Y_5\}$
- $f_{C_3} : (G \setminus \{g1, g3, g5\}) \times \{b, c\}$
- $Y_5 \in Max\{f_{C_3}(x) \ : \ x \in \{g2, g4, g6\}\}$
- $Y_5 \in Max\{f_{C_3}(g2), f_{C_3}(g4), f_{C_3}(g6)\} = Max\{\{c\}, \emptyset, \{c\}\} = \{c\}$

Hence:

- $C_5 = (X_5, Y_5) = (\{g1, g2, g3, g5, g6\}, \{c\})$

The Bordat algorithm

Direct subsumers of $C_4 = (X_4, Y_4) = (\{g2, g3, g5, g6\}, \{a, c, d\})$:

- $X_6 = X_4 \cup \{x \in G \setminus X_4 \ : \ f_{C_4}(x) = Y_6\}$
- $f_{C_4} : (G \setminus \{g2, g3, g5, g6\}) \times \{a, c, d\}$
- $Y_6 \in Max\{f_{C_4}(x) \ : \ x \in \{g1, g4\}$
- $Y_6 \in Max\{f_{C_4}(g1), f_{C_4}(g4)\}\} = Max\{\{c\}, \{a, d\}\}$

Hence:

- The first subsumer is identical to $C_5 = (\{g1, g2, g3, g5, g6\}, \{c\})$
- The second subsumer $C_6 = (\{g2, g3, g4, g5, g6\}, \{a, d\})$

Direct subsumers of $C_5 = (\{g1, g2, g3, g5, g6\}, \{c\})$:

- $X_7 = X_5 \cup \{x \in G \setminus X_5 \ : \ f_{C_5}(x) = Y_7\}$
- $f_{C_5} : (G \setminus \{g1, g2, g3, g5, g6\}) \times \{c\}$
- $Y_7 \in Max\{f_{C_5}(x) \ : \ x \in \{g4\}\} = Max\{f_{C_5}(g4)\}\} = \emptyset$

Hence:

- $C_7 = (\{g1, g2, g3, g4, g5, g6\}, \emptyset) = \top$

Similarly: the direct subsumer of $C_6 = (\{g2, g3, g4, g5, g6\}, \{a, d\})$ is C_7

Direct subsumers of $C_5 = (\{g1, g2, g3, g5, g6\}, \{c\})$:

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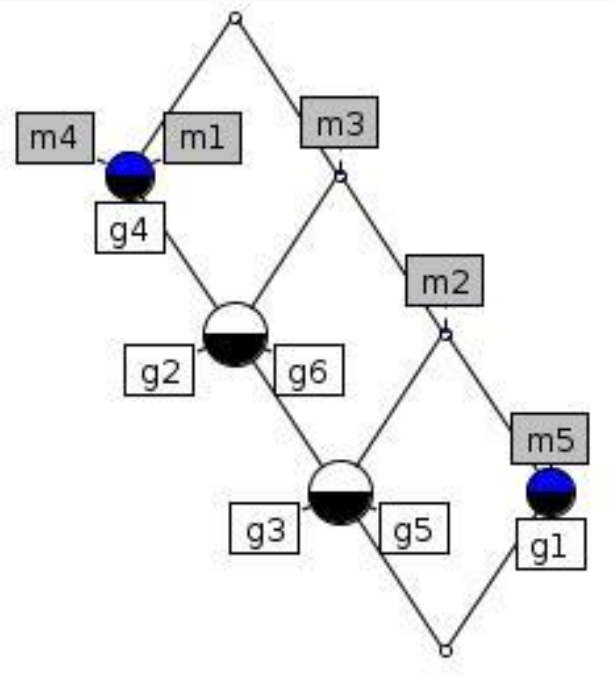
Hence:

- $C_7 = (\{g1, g2, g3, g4, g5, g6\}, \emptyset) = \top$

Similarly: the direct subsumer of $C_6 = (\{g2, g3, g4, g5, g6\}, \{a, d\})$ is C_7

The reduced labeling

G \ M	m1	m2	m3	m4	m5
g1		×	×		×
g2	×		×	×	
g3	×	×	×	×	
g4	×			×	
g5	×	×	×	×	
g6	×		×	×	



Reduced labeling: each object and each attribute appears only **once!**

Idea: attributes appear upwards (filters) **while** objects downwards (ideals).

Object and attribute concepts and reduced labelling...

The object concept of $g \in G$ is the concept $\gamma(g) = (\{g\}'', \{g\}')$. $\gamma(g)$ is the smallest concept containing g.

The attribute concept of $m \in M$ is the concept $\mu(m) = (\{m\}', \{m\}'')$. $\mu(m)$ is the biggest concept containing m.

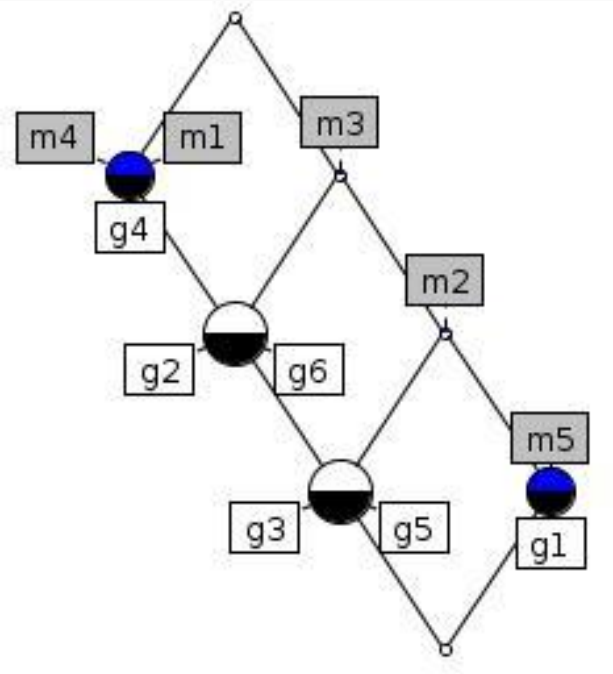
Reduced labeling: each object and attribute appears only once in the diagram.

The reduced labeling

G \ M	m1	m2	m3	m4	m5
g1		×	×		×
g2	×		×	×	
g3	×	×	×	×	
g4	×			×	
g5	×	×	×	×	
g6	×		×	×	

NB: For any concept (A, B),

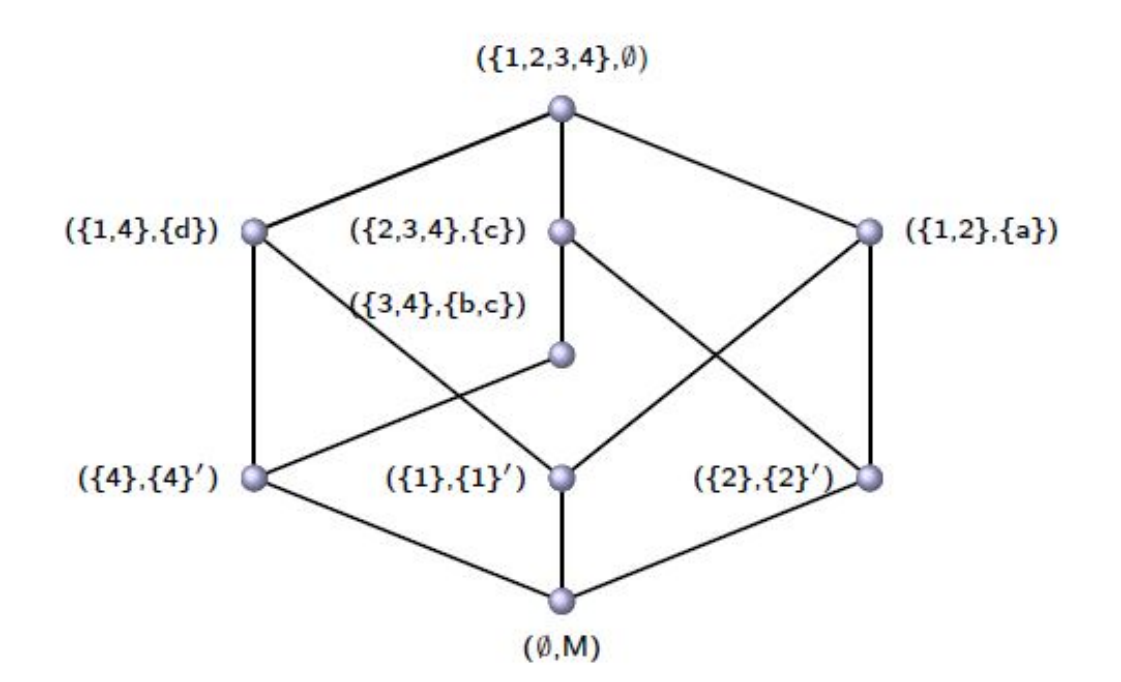
- $g \in A \iff \gamma(g) \leq (A, B)$
- $m \in B \iff (A, B) \leq \mu(m)$



NB: the attributes appear upwards (filters) and the objects downwards (ideals).

Exercise on 2D figures

	G \ M	a	b	c	d
1		×			×
2		×		×	
3			×	×	
4			×	×	×



From concept lattices to formal contexts

Let $(L, \leq)$ be a complete lattice. An element $x \in L$ is called

- $\vee$-irreducible if $x \neq \bigvee\{y \mid y < x\}$.
- $\wedge$-irreducible if $x \neq \bigwedge\{y \mid y > x\}$.

The sets of $\vee$- and $\wedge$-irreducibles of $(L, \leq)$ are denoted by $J(L)$ and $M(L)$, resp.

Corollary: For any lattice L, we have $L \cong \underline{\mathfrak{B}}(J(L), M(L), \leq)$.

Moreover: By interchanging G and M, we obtain the dual concept lattice $\underline{\mathfrak{B}}(G, M, I)^d \cong \underline{\mathfrak{B}}(M, G, I^{-1})$, through the anti-isomorphism $(B, A) \mapsto (A, B)$.

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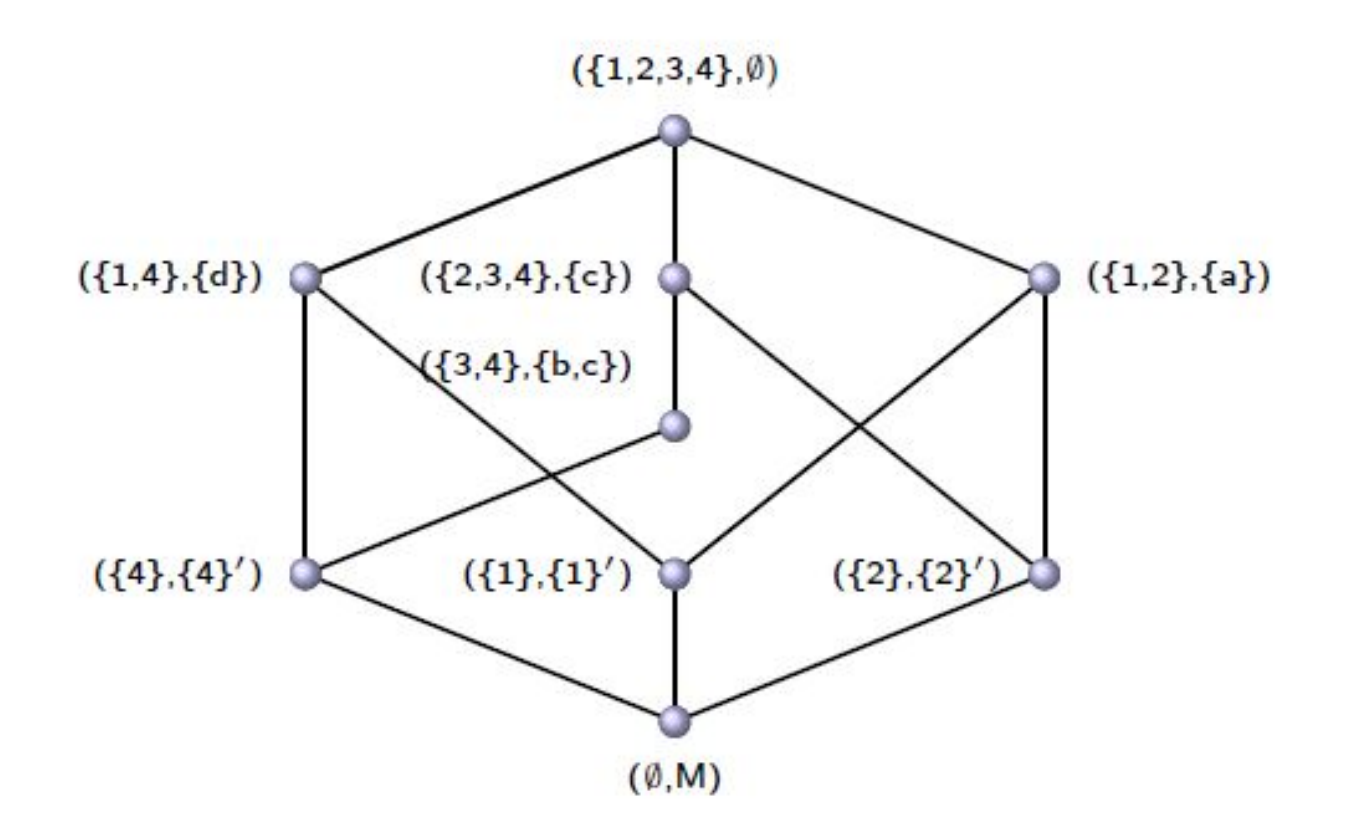
Back to the example...



	G \ M	a	b	c	d
1		×			×
2		×		×	
3			×	×	
4			×	×	×

a – has 3 vertices

b – has 4 vertices

c – has a right angle

d – equilateral

Reducing attribute sets

Let $\mathbb{K} = (G, M, I)$. Attribute $m \in M$ is **reducible** if $m' = G$ or

$$m' = \bigcap \{n' \mid n \in M \ \& \ n' \supset m'\}$$

NB: If m is reducible, **then**

$$\underline{\mathfrak{B}}(G, M, I) \cong \underline{\mathfrak{B}}(G, M \setminus \{m\}, I \cap (G \times (M \setminus \{m\})))$$

Example. Attribute m_k is reducible, since $m'_k = m'_i \cap m'_j$

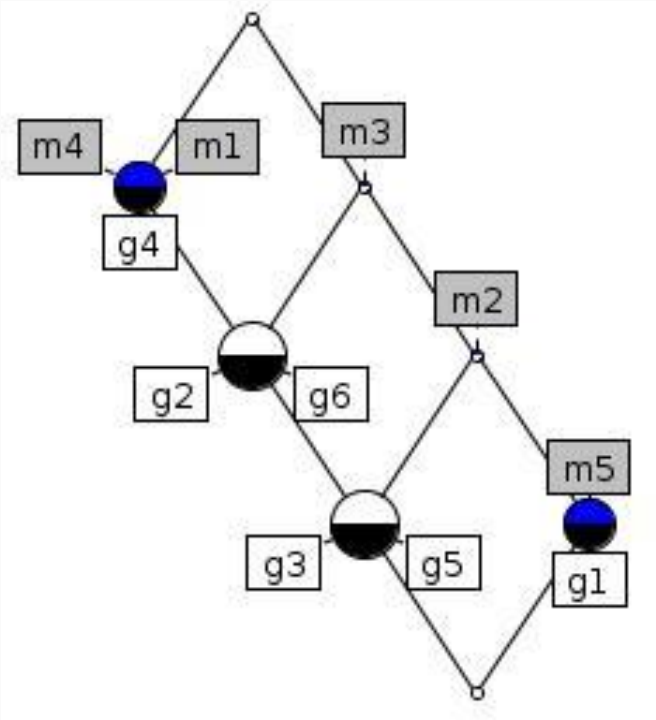
$G \setminus M$	$\dots$	m_i	$\dots$	m_j	$\dots$	m_k	$\dots$
g_1	$\dots$	$\times$	$\dots$		$\dots$		$\dots$
g_2	$\dots$	$\times$	$\dots$	$\times$	$\dots$	$\times$	$\dots$
g_3	$\dots$	$\times$	$\dots$	$\times$	$\dots$	$\times$	$\dots$
g_4	$\dots$		$\dots$	$\times$	$\dots$		$\dots$

Types of attributes

Introducing an attribute: an attribute α is introduced by a concept C when it is not present in any ascendant of C, i.e., the concept C corresponds to the attribute concept of α (the introducer of α).

Inheriting an attribute: an attribute α is inherited by a concept C when it is already present in a proper ascendant of C, i.e., C is subsumed by the introducer of α.

Types of attributes (examples)



$m3$ is introduced $(g12356, m3)$

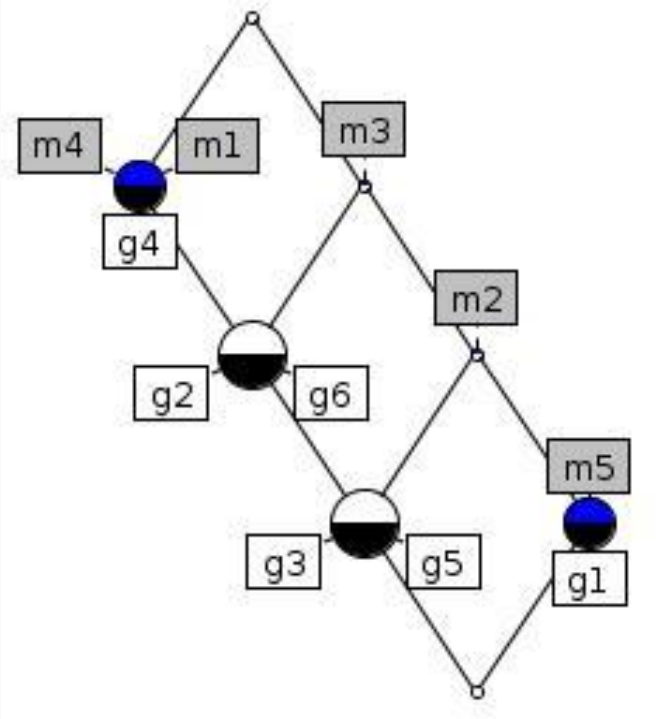
$m1$ and $m4$ are introduced by $(g23456, m14)$

$m2$ is introduced by $(g135, m23)$

$m3$ is inherited by $(g135, m23)$

$m1$, $m3$, and $m4$, are inherited by $(g2356, m134)$

Mutual implications between attributes having the same attribute-concept



Attributes having the same introducer are equivalent:

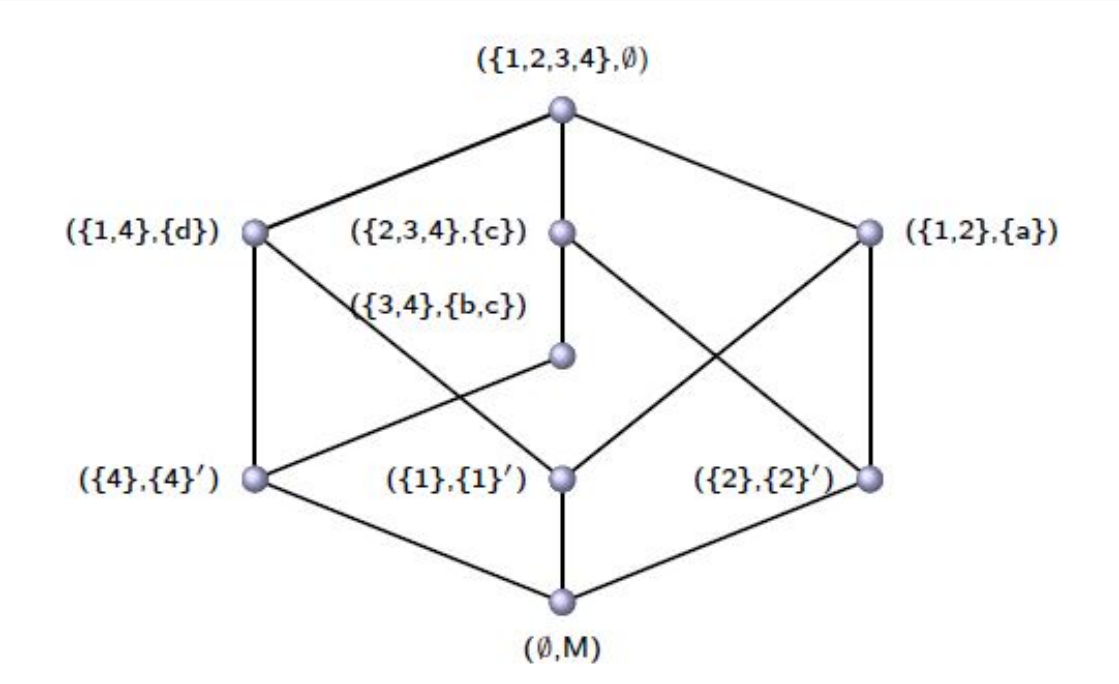
Ex: $m1 \longleftrightarrow m4$ for $(g23456, m14)$

Introduced attributes imply inherited attributes

If α is introduced, then it implies every inherited attribute in the introducer of α

Ex: $m2 \longrightarrow m3$ for $(g135, m23)$ and $m5 \longrightarrow m23$ for $(g1, m235)$

Exercise on 2D figures



	G \ M	a	b	c	d
1		×			×
2		×		×	
3			×	×	
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a – has 3 vertices

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Exercise: What rules that can be extracted directly from the concept lattice.

Many-valued contexts

A formal context is the basic data type of FCA.

However data are often in form of a **many-valued context** (rel. database)

A many-valued context is a quadruple (G, M, W, J), where W is the set of **attribute values**, and the **incidence relation** $J \subseteq G \times M \times W$ is such that $(g, m, w) \in J$, and $(g, m, v) \in J$ **implies** $w = v$.

G \ M	m1	m2	m3
g1	1	3	4
g2	2	2	3
g3	4	1	1
g4	2	2	1

An attribute m is complete if $\forall g \in G \exists w \in W$ such that $(g, m, w) \in J$

A many-valued context is complete if all its attributes are complete.

NB: For complete many-valued contexts the value of attribute m for object g is denoted by $m(g)$, i.e., $(g, m, m(g)) \in J$.

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Conceptual scaling

Many-valued contexts are translated a usual context via conceptual scaling. This is not automatic and some choices have to be made.

Examples of scalings:

- Nominal: $K = (N, N, =)$
- Ordinal: $K = (N, N, \leq)$
- Interordinal: $K = (N, N, \leq \cup \geq)$

A numerical example

G \ M	m1	m2	m3
g1	1	3	4
g2	2	2	3
g3	4	1	1
g4	2	2	1

Nominal Scaling:

G \ M	m1=1	m1=2	m1=4	m2=1	m2=2	m2=3	m3=1	m3=3	m3=4
g1	×					×			×
g2		×			×			×	
g3			×	×			×		
g4		×			×		×		

Interordinal Scaling:

G \ M	m1≤1	m1≥1	m1≤2	m1≥2	m1≤3	m1≥3	m1≤4	m1≥4	m2≤1	m2≥1
g1	×	×	×		×		×			×
g2		×	×	×	×		×			×
g3		×		×		×	×	×	×	×
g4		×	×	×	×		×			×

A refined algorithm to construct concept lattices:
`AllClosure` and `NextClosure`

The lectic order

Consider a linear ordering $<$ of M say $\quad M = \{a_1 < a_2 < \cdots < a_k\}$

Lectic order: $A \subseteq M$ is lectically smaller than $B \subseteq M$, $A < B$, if there is $i \in M$ such that:

$$i \in B \setminus A \;\; \textbf{and} \;\; \forall j < i \;\; \text{we have} \;\; j \in A \Leftrightarrow j \in B$$

Refinement: $A <_i B$ **iff** $i \in B \setminus A$ **and** $\forall j < i$ we have $j \in A \Leftrightarrow j \in B$

NB1: Take the characteristic vectors and consider the reversed lexic. order

NB2: If $A \subseteq B$, then $A < B$ (take the first in $B \setminus A$)

Example: Let $M = \{a < b < c < d < e < f\}$. Then $\{a, c, e\} < \{a, c, d\}$ **and**

$$\{a, c, e\} <_d \{a, c, d\} \;\; \textbf{but} \;\; \{a, c, e\} \not<_e \{a, c, d\}$$

The lectic order

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Refinement: $A <_i B$ **iff** $i \in B \setminus A$ **and** $\forall j < i$ we have $j \in A \Leftrightarrow j \in B$

NB1: Take the characteristic vectors and consider the reversed lexic. order

NB2: If $A \subseteq B$, then $A < B$ (take the first in $B \setminus A$)

Example: Let $M = \{a < b < c < d < e < f\}$. Then $\{a, c, e\} < \{a, c, d\}$ **and**

$$\{a, c, e\} <_d \{a, c, d\} \;\; \textbf{but} \; \{a, c, e\} \not<_e \{a, c, d\}$$

Closures in lectic order

NB3: For $A, B \subseteq M$, $A < B$ **iff** $A <_i B$, for some $i \in M$

NB4: For $A, B, C \subseteq M$, If $A <_i B$ **and** $A <_j C$ for $i < j$, **then** $C <_i B$

Let $A \subseteq M$ and $i \in M$. **Define**

$$A + i = ((A \cap \{j : j < i\}) \cup \{i\}) \quad \textbf{and} \quad A \oplus i = (A + i)''$$

Example: Let $A = \{a, c, d, f\}$ and $i = e$. **Then**

$$A \oplus e = ((\{a, c, d, f\} \cap \{a, b, c, d\}) \cup \{e\})'' = (\{a, c, d, e\})''$$

Theorem: The smallest concept intent larger than a given set $A \subset M$ with respect to the lectic order is $A \oplus i = (A + i)''$, with i being the largest element of M with $A <_i A \oplus i$

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Algorithms `AllClosure` and `NextClosure`

`AllClosure`: Generates the partially ordered set of closed attribute sets

Input: Closure operator $X \mapsto X''$ (formal context) on a set of attributes M

Output: All closed attribute sets in lectic order

1. $A \leftarrow \varnothing''$

2. While $A \neq M$ **do**

2.1. Output: A

2.2. $A \leftarrow \texttt{NextClosure}(A, M, '')$

Algorithm `NextClosure` (cont.)

`NextClosure`: Generates a the lectically next closed attribute set

Input: Closure operator $X \mapsto X''$ on a set of attributes M and $A \subseteq M$

Output: The lectically next closed attribute set, or M if it doesn't exist

1. For all $i \in M$ (in descending order) **do**

1.1. if $A <_i A \oplus i$ **then return** $A \oplus i$

2. return M

Exercise: compute the concept lattice of our example

	G \ M	a	b	c	d
1		×			×
2		×		×	
3			×	×	
4			×	×	×

Objects:

- **1** – equilateral triangle
- **2** – rectangle triangle
- **3** – rectangle
- **4** – square

Attributes:

- **a** – has 3 vertices
- **b** – has 4 vertices
- **c** – has a direct angle
- **d** – equilateral

Exercise: compute the concept lattice of our example

	G \ M	a	b	c	d
1		×			×
2		×		×	
3			×	×	
4			×	×	×

Tip: use a table for the computation!

A	i	$A + i$	$A \oplus i$	$A <_i A \oplus i$?	New intent?
$\varnothing$	d	$\{d\}$	$\{d\}'' = \{1, 4\}' = \{d\}$	yes	$\{d\}$
$\{d\}$	d	$\{d\}$	$\{d\}$	no	–
$\{d\}$	c	$\{c\}$	$\{c\}$	yes	$\{c\}$
$\{c\}$	d	$\{c, d\}$	$\{c, d\}'' = \{4\}' = \{b, c, d\}$	no	–
$\{c\}$	c	$\{c\}$	$\{c\}$	no	–
$\{c\}$	b	$\{b\}$	$\{b, c\}$	yes	$\{b, c\}$
$\{b, c\}$	d	$\{b, c, d\}$	$\{b, c, d\}$	yes	$\{b, c, d\}$
$\{b, c, d\}$	d	$\{b, c, d\}$	$\{b, c, d\}$	no	–
$\{b, c, d\}$	c	$\{b, c\}$	$\{b, c\}$	no	–
$\{b, c, d\}$	b	$\{b\}$	$\{b, c\}$	no	–
$\{b, c, d\}$	a	$\{a\}$	$\{a\}$	yes	$\{a\}$
$\{a\}$	d	$\{a, d\}$	$\{a, d\}$	yes	$\{a, d\}$
$\{a, d\}$	d	$\{a, d\}$	$\{a, d\}$	no	–
$\{a, d\}$	c	$\{a, c\}$	$\{a, c\}$	yes	$\{a, c\}$
$\{a, c\}$	d	$\{a, c, d\}$	$\{a, b, c, d\}$	no	–
$\{a, c\}$	c	$\{a, c\}$	$\{a, c\}$	no	–
$\{a, c\}$	b	$\{a, b\}$	$\{a, b, c, d\}$	yes	$\{a, b, c, d\}$

Some complexity issues

NB1: Consider context $\mathbb{K} = (G, G \neq)$ for an arbitrary finite set G. Then $\underline{\mathfrak{B}}(\mathbb{K})$ is isomorphic to the Boolean lattice 2^G with $2^{|G|}$ concepts.

NB2: Given a context $\mathbb{K} = (G, M, I)$, computing the size of $\underline{\mathfrak{B}}(\mathbb{K})$ is #P-complete [Kuznetsov 1989, 2001].

Intuition: #P is the class of counting problems associated with the decision problems in NP.

More formally: A problem is in #P if there is a non-deterministic, polynomial time Turing machine (TM) that, for each instance I of the problem, has a number of accepting computations that is exactly equal to the number of distinct solutions for instance I.

#P-complete: #P and #P-**hard**, i.e., any problem in #P can be reduced to it by a TM (or a #P-complete problem can be reduced to it).

NB: #P = P $\Rightarrow$ NP = P.

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NB: $\#\mathrm{P} = \mathrm{P} \Rightarrow \mathrm{NP} = \mathrm{P}$.

Some complexity mitigation

Good news: Computing concepts is possible with polynomial delay

Polynomial delay: If the time between the output of one solution and the next is bounded by a polynomial function of the input size (in the worst case)

Idea 1: Compute only concepts with big extents, i.e., large support (it can be done with polynomial delay)

Idea 2: Compute concepts with high stability or other quality indices.

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Exercises

Compute the concept lattices for each of the following formal contexts

1.

Objects/Attributes	1	2	3
A	X	X	X
B	X		X
C		X	X
D	X		

2.

model/functionality	**Mobile**	**Phone**	**Fax**	**Paper fax**
Sinus 14		X		
Nokia 6110	X	X		
T-Fax 301			X	X
T-Fax 360 PC				X

Implications & functional dependencies

Implications from concept lattices and implication models

Motivation: Mine attribute dependencies in cross-tables, i.e., implications!

Examples: implications represent data dependencies such as

- every number divisible by 2 and 3 is divisible by 6, or
- every patient with symptoms s_2 and s_5 also has symptoms s_1 and s_3.

Implication $A \to B$ holds in (G, M, I) if $A' \subseteq B'$, i.e., each object having all attributes from A also has all attributes from B.

Recall: For $A \to B$, $\bigwedge_{a \in A} \mu(a) \sqsubseteq \bigwedge_{b \in B} \mu(b)$.

Homework: frequent itemsets (itemset whose support is greater than a given threshold), `APriori` (mines implications based on frequent itemsets), ...

Def.: $T \subseteq M$ is a **model** of $A \to B$, denoted $T \models A \to B$, if $A \not\subseteq T$ or $B \subseteq T$.
If $\mathcal{I}$ set of implications, then $T \models \mathcal{I}$ if $T \models A \to B$, for every $A \to B \in \mathcal{I}$.
We denote the set of models of $\mathcal{I}$ by $Mod(\mathcal{I})$

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A basic algorithm to compute implication closures

Problem: For $X \subset M$, determine the $\mathcal{I}$-closure of X:

$$\mathcal{I}(X) = \bigcap \{B \colon X \subseteq B \subseteq M,\ B \models \mathcal{I}\}$$

NB: $\mathcal{I}(X)$ is the smallest model of $\mathcal{I}$ that contains X.

For $X \subset M$ and an implication set $\mathcal{I}$, **Define:**

$$X^{\mathcal{I}} = X \cup \bigcup \{B \colon A \to B \in \mathcal{I}, A \subseteq X\}$$

Result: $\mathcal{I}(X) = (((X^{\mathcal{I}})^{\mathcal{I}})^{\mathcal{I}})^{\cdots}$ and $\mathcal{I}(X)^{\mathcal{I}} = \mathcal{I}(X)$

`ImpClosure`: Computes the $\mathcal{I}$-closure of $X \subseteq M$ (repeated until $X^{\mathcal{I}} = X$)

Repeat: For all $A \to B \in \mathcal{I}$ **do**

  if $A \subseteq X$ **then** $X \leftarrow X \cup B, \quad \mathcal{I} \leftarrow \mathcal{I} \setminus \{A \to B\}$

Return: X

However: Quadratic in the number of implications...

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However: Quadratic in the number of implications...

Implications from a logical perspective: bases

Implications satisfy **Armstrong rules**:

$$\frac{}{X \to X} \quad , \quad \frac{X \to Y}{X \cup Z \to Y} \quad , \quad \frac{X \to Y, Y \cup Z \to W}{X \cup Z \to W},$$

An **implication cover** is a subset of implications from which all other implications can be derived by means of Armstrong rules.

An **implication base** is a minimal (by inclusion) implication cover.

Recall: $\mathcal{I} \models A \to B$ (semantic), $\mathcal{I} \vdash A \to B$ (syntactic), and coincide!

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Exercises I

Consider the following attributes: normal blood pressure (nbp), high blood pressure (hbp), watches TV (TV), eats unhealthy food (uf), runs regularly (r).

-	**nbp**	**hbp**	**TV**	**uf**	**r**
A	X				X
B	X			X	X
C		X	X	X	
D		X		X	
E	X				

Check whether the following implications are valid: $\{r\} \rightarrow \{nbp\}$, $\{TV, uf\} \rightarrow \{hbp\}$, $\{TV\} \rightarrow \{hbp\}$, $\{uf\} \rightarrow \{hbp\}$, $\{nbp\} \rightarrow \{r\}$, $\{nbp,hbp\} \rightarrow \{r,TV\}$, $\{uf,r\} \rightarrow \{r\}$

Exercises II

Consider the following theories:

- $T_1 = \{\{y_3\} \to \{y_1, y_2\}, \{y_1, y_3\} \to \{y_2\}\}$.
- $T_2 = \{\{y_3\} \to \{y_1, y_2\}\}$.
- $T_3 = \{\{y_1, y_3\} \to \{y_2\}\}$.
- $T_4 = \{\{y1\} \to \{y_3\}, \{y_3\} \to \{y_1\}, \{y_2\} \to \{y_2\}\}$.
- $T_5 = \varnothing$.
- $T_6 = \{\varnothing \to \{y_1\}, \varnothing \to \{y_3\}\}$.
- $T_7 = \{\{y_1\} \to \varnothing, \{y_2\} \to \varnothing, \{y3\} \to \varnothing\}$.
- $T_8 = \{\{y_1\} \to \{y_2\}, \{y_2\} \to \{y_3\}, \{y_3\} \to \{y_1\}\}$.

Compute the corresponding sets of models.

Exercises III

1. Consider the following theories T together with an implication $A \to B$ (over $M = \{y_1, y_2, y_3\}$). Decide whether $T \models A \to B$
   - $T = \{\{y_3\} \to \{y_1, y_2\}, \{y_1, y_3\} \to \{y_2\}\}$, where $A \to B$ is $\{y_2, y_3\} \to \{y_1\}$.
   - $T = \{\{y_3\} \to \{y_1, y_2\}, \{y_1, y_3\} \to \{y_2\}\}$, where $A \to B$ is $\{y_2\} \to \{y_1\}$.

2. Let $T = \{\{y_1, y_4\} \to \{y_3\}, \{y_2, y_4\} \to \{y_1\}, \{y_1, y_2\} \to \{y_4, y_7\}, \{y_2, y_7\} \to \{y_3\}, \{y_6\} \to \{y_4\}, \{y_2, y_8\} \to \{y_3\}, \{y_9\} \to \{y_1, y_2, y_7\}\}$ (over $M = \{y_1, \ldots, y_{10}\}$). Decide whether $T \models A \to B$ when $A \to B$ is
   - $\{y_2, y_5, y_6\} \to \{y_3, y_7\}$.
   - $\{y_1, y_2, y_8\} \to \{y_4, y_7\}$.

Non-redundant bases of implications

A set $\mathcal{I}$ of implications is called non-redundant if for any $A \to B \in \mathcal{I}$,

$$\mathcal{I} \setminus \{A \to B\} \not\models A \to B.$$

To remove redundancy we can use:

Repeat: For all $A \to B \in \mathcal{I}$ **do**

$\mathcal{I}' = \mathcal{I} \setminus \{A \to B\}$
if $\mathcal{I}' \models A \to B$ **then** $\mathcal{I} = \mathcal{I}'$

Return: $\mathcal{I}$

NB1: An **(implication) basis** is a non-redundant cover (complete)

NB2: $\mathcal{I}$ is a cover of (G, M, I) iff $Mod(\mathcal{I})$ are exactly the intents of $\underline{\mathfrak{B}}(G, M, I)$

NB: Any basis $\mathcal{I}$ can be seen as a "partial" closure operator!

Question: How to compute X'', for any $X \subseteq M$?

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Generator implication cover

$D \subseteq M$ is a **generator** of a closed $B = B'' \subseteq M$, if $D \subseteq B$ and $D'' = B$.

$D \subseteq M$ is a **minimal generator** (key) if for all $E \subset D$, $\;E'' \neq D'' = B''$.

It is called **nontrivial** if $D \neq D'' = B''$. The set of all nontrivial minimal generators of B is denoted by $\text{mingen}(B)$.

Generator implication cover: $\{F \to (F'' \setminus F) \mid F \subseteq M,\, F \in \text{mingen}(F'')\}$.

Direct (proper premise) basis

$A \subseteq M$ is a **proper premise** if

$$A'' \neq A \cup \bigcup_{n \in A} (A \setminus \{n\})''.$$

NB1: The closure of a proper premise cannot be deduced from the union of closures of proper subsets of A.

NB2: Every proper premise is a minimal generator, but not vice versa.

Direct basis[1]**:** $\mathcal{D} = \{A \to (A'' \setminus A) \mid A \text{ is a proper premise}\}$

Or: $\mathcal{D} = \{A \to A'' \mid A \text{ is a proper premise}\}$

[1] Also: "Proper premise base", "Direct Canonical Basis", "Karell Basis", from K. Bertet, B. Monjardet. The multiple facets of the canonical direct unit implicational basis, Theoretical Computer Science, Vol. 411, Issues 22-24, 2010, Pages 2155-2166, ISSN 0304-3975, https://doi.org/10.1016/j.tcs.2009.12.021.

Minimum implication (canonical) basis

$P \subseteq M$ is a **pseudo-intent** (or **pseudo-closed**) if it is not an intent ($P \neq P''$), and for any pseudo-intent $Q \subset P$, we have $Q'' \subset P$.

Duquenne-Guigues (canonical) basis:

$$\mathcal{I} = \{P \to (P'' \setminus P) \mid P \text{ pseudo-intent } \}.$$

Or: $\mathcal{I} = \{P \to P'' \mid P \text{ pseudo-intent}\}$.

Theorem: The D-G basis is a (cardinality) minimal implication basis.

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Exercises IV

Recall Exercise I: normal blood pressure (nbp), high blood pressure (hbp), watches TV (TV), eats unhealthy food (uf), runs regularly (r).

-	**nbp**	**hbp**	**TV**	**uf**	**r**
A	X				X
B	X			X	X
C		X	X	X	
D		X		X	
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Check whether the following theories are complete:

1. $T = \{\{\text{r}\} \to \{\text{nbp}\}, \{\text{TV, uf}\} \to \{\text{hbp}\}, \{\text{r,uf}\} \to \{\text{TV}\}$.
2. $T = \{\{\text{r}\} \to \{\text{nbp}\}, \{\text{TV, uf}\} \to \{\text{hbp}\}, \{\text{TV}\} \to \{\text{uf}\}, \{\text{TV}\} \to \{\text{hbp}\}\}$.
3. $T = \{\{\text{r}\} \to \{\text{nbp}\}, \{\text{TV, uf}\} \to \{\text{hbp}\}, \{\text{TV}\} \to \{\text{uf}\},$
   $\{\text{TV}\} \to \{\text{hbp}\}, \{\text{hbp,TV}\} \to \{\text{uf}\}, \{\text{nbp,uf}\} \to \{\text{r}\}, \{\text{hbp}\} \to \{\text{uf}\},$
   $\{\text{uf,r}\} \to \{\text{nbp}\}, \{\text{nbp,TV}\} \to \{\text{r}\}, \{\text{hbp,nbp}\} \to \{\text{r,TV}\}\}$.

Exercises IV (cont.)

Recall Exercise I: normal blood pressure (nbp), high blood pressure (hbp), watches TV (TV), eats unhealthy food (uf), runs regularly (r).

-	**nbp**	**hbp**	**TV**	**uf**	**r**
A	X				X
B	X			X	X
C		X	X	X	
D		X		X	
E	X				

Reduce the following theory into a non-redundant one.

$$T = \{\{\text{r}\} \to \{\text{nbp}\}, \{\text{TV, uf}\} \to \{\text{hbp}\}, \{\text{TV}\} \to \{\text{uf}\},$$
$$\{\text{TV}\} \to \{\text{hbp}\}, \{\text{hbp,TV}\} \to \{\text{uf}\}, \{\text{nbp,uf}\} \to \{\text{r}\}, \{\text{hbp}\} \to \{\text{uf}\},$$
$$\{\text{uf,r}\} \to \{\text{nbp}\}, \{\text{nbp,TV}\} \to \{\text{r}\}, \{\text{hbp,nbp}\} \to \{\text{r,TV}\}\}.$$

Some useful links

More to come but...

- https://latviz.loria.fr/
- https://upriss.github.io/fca/fca.html
- Binary data: https://github.com/smartfca/caspailleur/
- Complex patterns with FCA (Part II): https://smartfca.github.io/paspailleur/

Functional dependency: For many-valued attributes X, Y in complete many-valued context (G, M, W, J), Y is **functionally dependent** of X, $X \rightarrow_{fd} Y$, if for every pair $g, h \in G$:

$$(\forall m \in X \ m(g) = m(h)) \Rightarrow (\forall n \in Y \ n(g) = n(h)).$$

Example: Many-valued context $K = (G, M, W, J)$, where $b \rightarrow_{fd} c$ is **valid**

G \ M	a	b	c
1	a_1	b_1	c_1
2	a_2	b_1	c_1
3	a_1	b_2	c_2
4	a_2	b_3	c_2

Example: Many-valued context $K = (G, M, W, J)$, where $b \rightarrow_{fd} c$ is **not valid**

G \ M	a	b	c
1	a_1	b_1	c_1
2	a_2	b_1	c_2
3	a_1	b_2	c_2
4	a_2	b_3	c_2

Functional Dependencies as Implications

Proposition: For $\mathbb{K} = (G, M, W, J)$, consider $\mathbb{K}_N := (\mathcal{P}_2(G), M, I)$ where $\mathcal{P}_2(G)$ is the set of pairs of different objects from G, and I is defined by

$$\{g, h\}\, I\, m \Leftrightarrow m(g) = m(h).$$

Then $X \rightarrow_{fd} Y$ holds in $\mathbb{K}$ **iff** the implication $X \rightarrow Y$ holds in $\mathbb{K}_N$.

Example: Consider the following many-valued context $\mathbb{K}$:

G \ M	a	b	c	d
1	α	α	β	β
2	α	γ	α	α
3	δ	α	δ	α
4	α	γ	α	δ

and the following formal context $\mathbb{K}_N = (G, M, I)$:

G \ M	a	b	c	d
{1,2}	×			
{1,3}		×		
{1,4}	×			
{2,3}				×
{2,4}	×	×	×	
{3,4}				

NB: The D-G basis of $\mathbb{K}_N = (G, M, I)$ is the minimal cover of FDs of $\mathbb{K}$.

Conversely: For any $\mathbb{K} = (G, M, I)$, there is $\mathbb{K}_W = (G, M, W, J)$ such that...

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Association rules & hypotheses

Association rules

Idea: Association rules as "partial" FCA implications [Agrawal et al. 90's]

More precisely: An **association rule** of (G, M, I) is an exp. $A \to_{c,s} B$, where

- $c = \frac{|(A \cup B)'|}{|A'|} \in [0, 1]$, called **confidence**, $\mathrm{conf}(A \to B)$;
- $s = \frac{|(A \cup B)'|}{|G|} \in [0, 1]$, called **support**, $\mathrm{supp}(A \to B)$.

Further metrics of interest: *Lift*

$$\mathrm{lift}(A \to B) = \frac{\mathrm{conf}(A \to B)}{\mathrm{supp}(B)}$$

Question: What is a minimal representation of the set of association rules, from which all association rules can be obtained using "admissible transformations"?

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Question: What is a minimal representation of the set of association rules, from which all association rules can be obtained using "admissible transformations"?

Covers of association rules

Idea: Consider $A \rightarrow_{c,s} B$. Under fixed c and support s we try to **reduce** its premise and **increase** its conclusion

1. Decreasing premises: For fixed c and s we can decrease the premise to $D \subseteq A$ such that $(D \cup B)' = (A \cup B)' = A' \cap B' = D' \cap B'$, **i.e.**, $D' = A' = A'''$. **Thus:** minimal D is by definition a minimal generator of A'', i.e. $D \in \text{mingen}(A'')$.

Recall 1: $D \subseteq M$ is a **generator** of $B = B'' \subseteq M$ if $D \subseteq B$ and $D'' = B = B''$.

Recall 2: $D \subseteq M$ is a **minimal generator** if for every $E \subset D$, $E'' \neq D'' = B''$.

Covers of association rules (cont.)

2. Increasing conclusion. Conclusion B can be increased by Δ such that $(A \cup B)' = (A \cup B \cup \Delta)' = (A \cup B)' \cap \Delta'$, which is possible **only when** $(A \cup B)' \subseteq \Delta'$.

Or equiv.: to $A \cup B \to \Delta$ and to $\Delta \subseteq (A \cup B)''$. **Thus:** the conclusion can be increased up to $(A \cup B)''$.

Exercise: Show that these transformations do not decrease confidence nor support!

Conclusion 1: The set $CP(K) = \{D \to (A \cup B)'' \mid D \in \text{ mingen } (A'')\}$ is a cover of the set of all association rules.

Conclusion 2: All other rules can be otained by admissible transformations on rules from $CP(K)$, **i.e.**, $CP(K)$ is a cover of association rules.

Covers of association rules (cont.)

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Bases of association rules

Consider an assoc. rule of the form $D \to (A \cup B)''$, where $D \in \text{mingen}(A'')$.
In the concept lattice this rule corresponds to a path (A', A'') to the concept $((A \cup B)', (A \cup B)'')$.

NB: If (A', A'') is not an upper cover $((A \cup B)', (A \cup B)'')$, **then** there is a (E', E'') such that $(A', A'') \succ (E', E'') \succ ((A \cup B)', (A \cup B)'')$.

Consider $D \to E''$ ($D \in \text{mingen}(A'')$), and $F \to (A \cup B)''$ ($F \in \text{mingen}(E'')$).
Then

1. the confidence of the first rule is $c_1 = \frac{|E'|}{|A'|}$,
2. the confidence of the second rule is $c_2 = \frac{|(A \cup B)'|}{|E'|}$, and
3. the confidence of $D \to (A \cup B)''$, where $D \in \text{mingen}(A'')$ is

$$c = \frac{|(A \cup B)'|}{|A'|} = \frac{|E'|}{|A'|} \cdot \frac{|(A \cup B)'|}{|E'|} = c_1 \cdot c_2.$$

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Bases of association rules (cont.)

Hence the cover of the set of association rules can be made even smaller by restricting it to

$$\mathcal{A} = \{F \to (E'' \setminus F'') \mid F \subseteq M, F \in \text{mingen}(F''), \textbf{ and } (F', F'') \succ (E', E'')\}$$

NB1: They correspond to the arcs of the lattice diagram!

NB2: Support and confidence of other rules from the cover can be obtained by multiplying supports along the respective paths in the lattice

NB3: To minimize the cover of association rules, i.e., a **basis**, keep only those rules from $CP(K)$ that correspond to edges in a spanning tree of the lattice

Bases of association rules (cont.)

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Recall: Finding association rules...

Task: Find all "frequent" ($s >$threshold) association rules with $c > \theta$.

Solution steps:

1. Find all frequent "closed itemsets" (frequent intents)
2. For each frequent intent B find all its maximal subintents $A_1, \ldots, A_n$
3. Retain only those A_i for which $\text{conf}(A_i \to B) \geq \theta$
4. Find minimal generators of the remaining A_i, compose rules of the form $mingen(A_i) \to B$.

Luxenburger basis:

- Spanning tree of the concept lattice diagram
  (confident rules btw neigh. closed intents, w.r.t. supp. & conf.)
- Duquenne-Guigues implication base

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PART II

Recall the Formal Concept Analysis

What the course is about

The course "Symbolic Knowledge Discovery" raises three questions:

1. What is "Symbolic Knowledge"? $\Longrightarrow$ Something human-readable: concepts, implications, associations, Knowledge Graphs and Ontologies
2. How to "discover" it? $\Longrightarrow$ FCA, Frequent Itemset mining, association rule mining, etc
3. What for? $\Longrightarrow$ To learn something new, to make AI speak human language

What is Knowledge? What is thinking?

The year is 1662. Two monks Antoine Arnauld and Pierre Nicole[2] think about Thinking and publish a book *La logique, ou l'art de penser* also known as *Logique de Port-Royal*.

Among many things, they talk about:

- *substances* of real world that are always singular
- *attributes* that are unchanging aspects of substances
- *ideas* that group multiple substances together

They propose that the ideas have two important aspects:

- *comprehension*, i.e. the set of attributes essential to the idea,
- *extension*, i.e. the particular objects to which the idea applies.

With time, the term substance transitioned into *object*, and comprehension became *intention*.

More on the topic: `https://plato.stanford.edu/entries/port-royal-logic/`

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A more recent history

The year is 1832. Évariste Galois – a rebelious student of 20 years old – solves the problem of whether polynomials of 5th degree can be solved by radicals. But he skips some classical student books of the time so 1) he generates ideas not like the others, and 2) nobody understands what does he write[3].

On May 30 he fights in a duel and dies. Two days before he collects about 60 pages of his works and sends them to his friend with a note: *"Tu prieras publiquement Jacobi ou Gauss de donner leur avis, non sur la vérité, mais sur l'importance des théorèmes. Après cela, il y aura, j'espère, des gens qui trouveront leur profit à déchiffrer tout ce gâchis."*

In 1842 Joseph Liouville starts deciphering the notes of Galois and publishes them in 1846. The foundings eventualy lead to the foundation of *group theory*, *Galois theory* including *Galois connections*, and quite the whole of *modern algebra*. Some say, it is then, when the mathematics became complicated.

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An even more recent history

The year is 1940. Professor Garett Birkhoff teaches in Harvard and publishes his monograph *Lattice Theory*. The latter describes lattices, that are algebraic structures consisting of a set equipped with supremum $\vee$ and infimum $\wedge$ operations.

Lattice theory naturally extends Group theory and relates to Boolean algebra, to partially ordered sets, to geometry, and to many many other branches of mathematics.

Among others, the notion of a lattice gives rise to:

- distributive lattices,
- complete lattices,
- join-irreducible and meet-irreducible elements,
- join and meet semilattices.

In 1966 G. Birkhoff makes a talk on a symposium on *Trends in Lattice*. The printed version of the talk spans 38 pages and finishes with sentence *"I hope I have made a convincing case for the thesis that lattices can do things for you, no matter what kind of mathematician you are"*.

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The year is 1982. Professor Rudolf Wille works in Darmstadt and publishes the paper "Restructuring lattice theory: an approach based on hierarchies of concepts". He and his research group (including Bernard Ganter) tries to figure if lattices can be useful for someone who is not a mathematician.

The go through a lot of philosophical books and find out that:

- If you take a binary table with objects G as rows and attributes M as columns ,
- find all the the ways $\mathbb{A}$ to group objects w.r.t. attributes,
- find the full descriptions $\mathbb{B}$ for every group of objects,
- then both $\mathbb{A}$ and $\mathbb{B}$ would be complete lattices, isomorphic to each other,
- by combining two lattices together we obtain the concept lattice $\mathbb{L}$ where each element represents an *idea* defined by its extension A and its intention B,
- Moreover, the operations that we use to describe objects with attributes and attributes with objects are *Galois connections*!

So all 300 years of history coincide in one moment in time in one theory for data analysis.

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The go through a lot of philosophical books and find out that:

- If you take a binary table with objects G as rows and attributes M as columns ,
- find all the the ways $\mathbb{A}$ to group objects w.r.t. attributes,
- find the full descriptions $\mathbb{B}$ for every group of objects,
- then both $\mathbb{A}$ and $\mathbb{B}$ would be complete lattices, isomorphic to each other,
- by combining two lattices together we obtain the concept lattice $\mathbb{L}$ where each element represents an *idea* defined by its extension A and its intention B,
- Moreover, the operations that we use to describe objects with attributes and attributes with objects are *Galois connections*!

So all 300 years of history coincide in one moment in time in one theory for data analysis.

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Honourable mentions

There are many more people who made Formal Concept Analysis possible. Here are just some of them:

- Bernard Monjardet,
- Marc Krasner,
- Nicolas Pippenger,
- Vincent Duquenne,
- and many many many others...

Pros and Cons of FCA

Formal Concept Analysis has many advantages:

- A solid mathematical basis in Lattice Theory
- A solid philosophical background in Logique de Port Royale
- Is targeted specifically to data analysis
- It touches many areas of modern Symbolic AI
- Its mathematical language is fairly simple (though, unusual)

But there is one disadvantage: *FCA does not scale*

- It only applies to binary data,
- Its algorithms often explode when dealing with more than 50 binary columns,
- It focuses a lot on maximal descriptions (i.e. intents) that tend to be big (by definition) and thus unreadable.

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Plan for the day

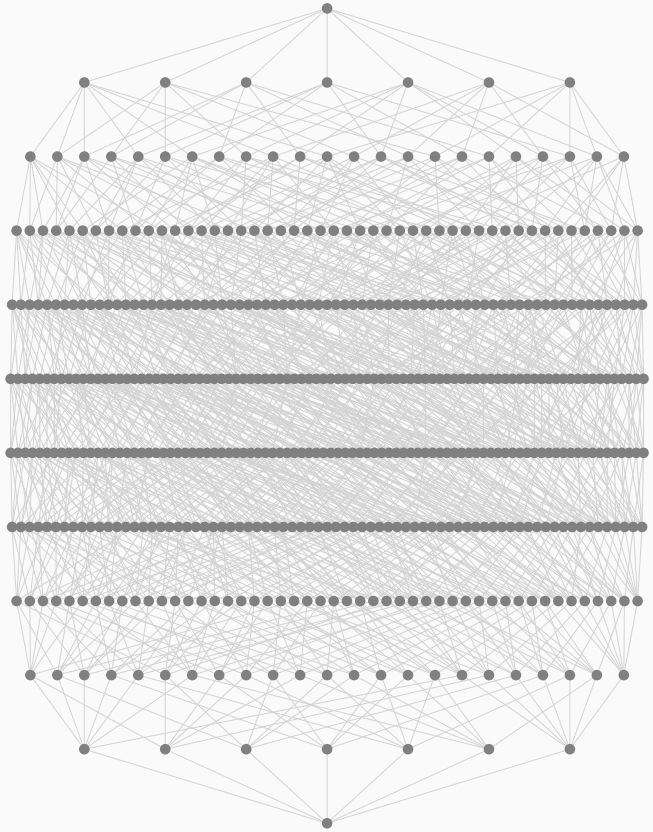
How to make FCA work for big real data?

- Simplify the task by mining only a few *interesting* concepts $\Longrightarrow$ CM Topic 1
- Complexify the model by using more complex formalisms $\Longrightarrow$ CM Topic 2
- Combine the two approaches within a software $\Longrightarrow$ TD

Interesting descriptions

The explosive problem

All descriptions of the Titanic data (with some discretisation)



Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

$$|L| \leq 2^{|M|}.$$

The explosive problem

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Solution Mine only *interesting* concepts and descriptions w.r.t. their support.

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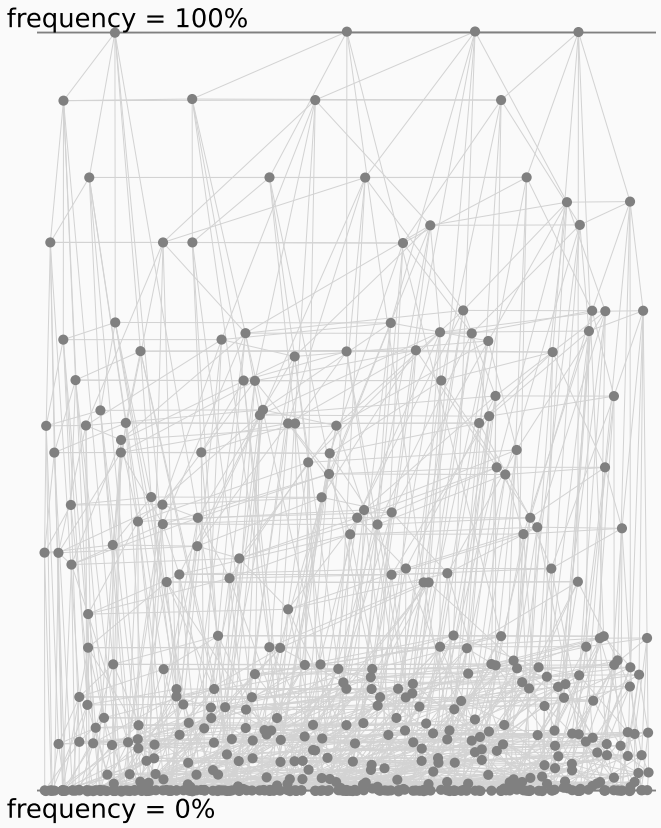


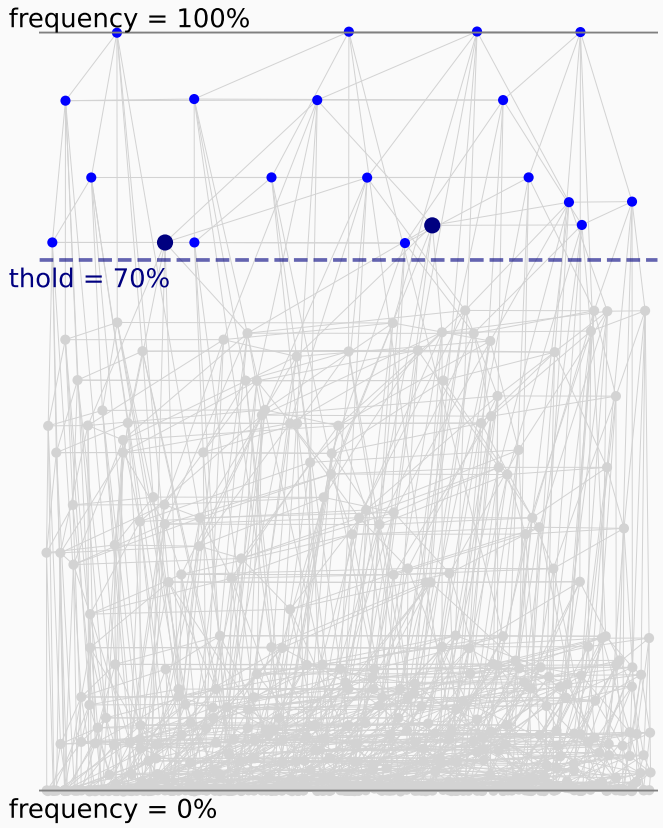
Frequent descriptions

Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

$$|L| \leq 2^{|M|}.$$

Solution Mine only *frequent* concepts and descriptions at the top of the lattice.

Frequent descriptions



Frequent descriptions (cont.)

Support Recall that *support* of a description $D \subseteq M$ is the number of objects that it describes:

$$\forall D \subseteq M,\ \mathtt{supp}(D) = |\{g \in G \mid D \subseteq g'\}| \tag{1}$$

Frequent description Given a threshold $\mathtt{min_supp} \in \mathbb{N}$, description $D \subseteq M$ is called *frequent* if it covers more than $\mathtt{min_supp}$ objects:

$$\forall D \subseteq M,\ D \text{ is frequent } \iff \mathtt{supp}(D) \geq \mathtt{min_supp}. \tag{2}$$

Maximal frequent description Given a threshold $\mathtt{min_supp} \in \mathbb{N}$, description $D \subseteq M$ is called *maximal frequent* if it covers more than $\mathtt{min_supp}$ objects and all greater descriptions cover less than $\mathtt{min_supp}$ objects:

$$\begin{aligned}\forall D \subseteq M,\ D \text{ is max. frequent } \iff & \mathtt{supp}(D) \geq \mathtt{min_supp} \text{ and} \\ & \mathtt{supp}(E) < \mathtt{min_supp},\ \forall E, D \subset E \subseteq M.\end{aligned} \tag{3}$$

Here comes FCA Every maximal frequent description is the intent of a formal concept!

Mining maximal frequent descriptions

Antimonotonic property The size of a description is antimonotonic to its frequency. That is, the bigger is the description, the less frequent it is:

$$\forall D, E \subseteq M,\ D \subseteq E \implies \mathtt{supp}(D) \geq \mathtt{supp}(E). \tag{4}$$

So, the generic algorithm for mining frequent descriptions would be:

- Put the empty description into a queue $Q = \langle \emptyset \rangle$
- While the queue is not empty ($|Q| > 0$):
  - Take description from the queue $D = Q.\mathtt{pop}()$
  - If D is frequent:
    - Add next descriptions to the queue: $\forall m \in M \setminus D,\ Q.\mathtt{add}(D \cup \{m\})$

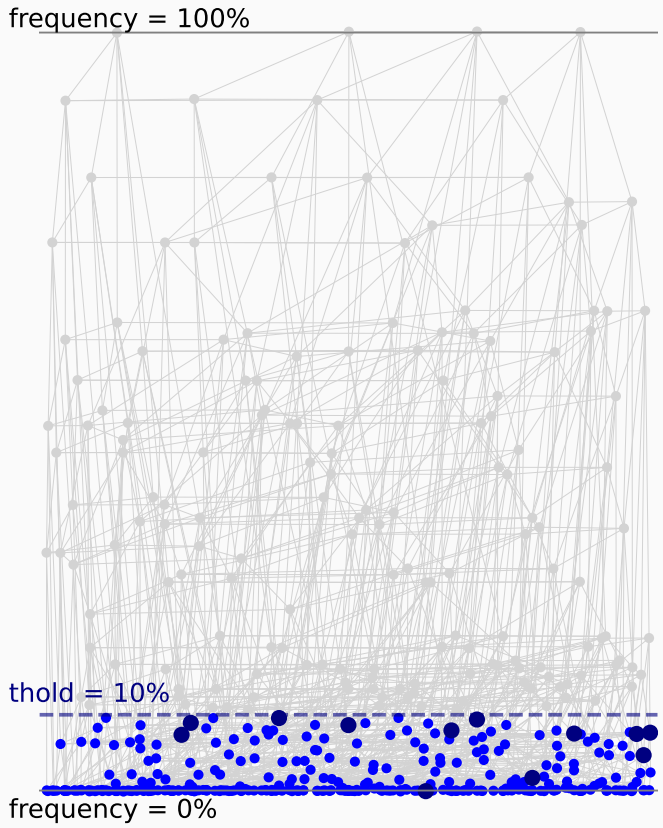
Rare descriptions

Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

$$|L| \leq 2^{|M|}.$$

Solution Mine only *rare* concepts and descriptions at the bottom of the lattice.

Rare descriptions



Rare descriptions (cont.)

Rare description Given a threshold $\texttt{max_supp} \in \mathbb{N}$, description $D \subseteq M$ is called *rare* if it covers less than $\texttt{max_supp}$ objects:

$$\forall D \subseteq M,\ D \text{ is rare} \iff \texttt{supp}(D) < \texttt{max_supp}. \tag{5}$$

Minimal rare description Given a threshold $\texttt{max_supp} \in \mathbb{N}$, description $D \subseteq M$ is called *minimal rare* if it covers less than $\texttt{max_supp}$ objects and all smaller descriptions cover more than $\texttt{max_supp}$ objects:

$$\begin{aligned} \forall D \subseteq M,\ D \text{ is min. rare} \iff & \texttt{supp}(D) < \texttt{max_supp} \text{ and} \\ & \texttt{supp}(E) \geq \texttt{max_supp},\ \forall E \subset D. \end{aligned} \tag{6}$$

Here comes FCA Every minimal rare description is a minimal generator of a formal concept!

Algorithm for mining minimal rare descriptions

SubKey property Every subset of a minimal generator is a minimal generator:

$$\forall D,\ D \text{ is a min. generator} \implies E \text{ is a min. generator},\ \forall E \subset D. \tag{7}$$

ProtoAlgo So, the generic algorithm for mining rare descriptions would be:

- Put the empty description into a queue $Q = \langle \emptyset \rangle$
- While the queue is not empty ($|Q| > 0$):
  - Take description from the queue $D = Q.\mathrm{pop}()$
  - Test if D is a minimal generator, i.e. every smaller $E \subset D$ describes more objects
  - If D is minimal, evaluate its support
  - If D is minimal and not rare, add next descriptions to the queue $\forall m \in M \setminus D,\ Q.\mathrm{add}(D \cup \{m\})$

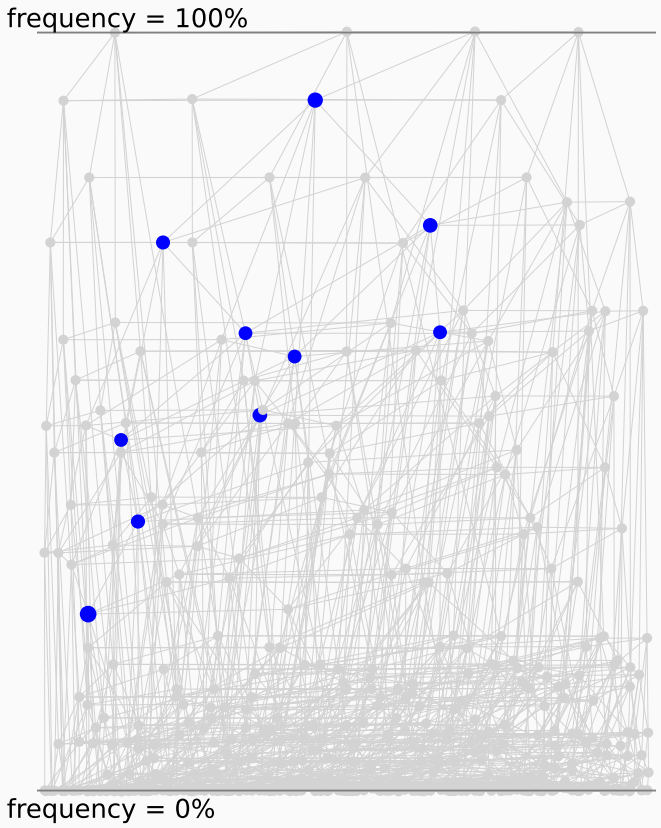
Δ-stable descriptions

Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

$$|L| \leq 2^{|M|}.$$

Solution Mine only *stable* concepts and descriptions that lie far from their super-concepts.

Δ-stable concepts



Δ-Stable descriptions (cont.)

Δ-Stability Δ-*stability* of description $D \subseteq M$ is the minimal difference between the support of the description and any of its superdescriptions:

$$\forall D \subseteq M,\ \Delta(D) = \min_{D \subset E \subseteq M} \big(\mathrm{supp}(D) - \mathrm{supp}(E)\big) = \mathrm{supp}(D) - \max_{D \subset E \subseteq M} \mathrm{supp}(E) \tag{8}$$

Δ-Stable description Given a threshold $\Delta_{\min} \in \mathbb{N}$, description $D \subseteq M$ is called $\Delta_{\min}$-*stable* if its Δ-stability is at least $\Delta_{\min}$:

$$\forall D \subseteq M,\ D \text{ is } \Delta_{\min}\text{-stable} \iff \Delta(D) \geq \Delta_{\min}. \tag{9}$$

Here comes FCA Every non-closed (i.e. non-maximal) description has zero Δ-stability. So only concept intents are stable!

Algorithm for mining Δ-stable descriptions

Projection-antimonotonicity property Let the set of attributes be lectically ordered $M = \{m_1, m_2, \ldots, m_n\}$ and let $i, j \in \mathbb{R}, i \leq j$ be two indices of attributes. Then Δ-stability computed w.r.t. the first i attributes is never smaller than the Δ-stability computed w.r.t. the first j attributes. That is, the more attributes we consider, the less stable the description becomes:

$$\begin{aligned} &\text{Let } M_i = \{m_1, m_2, \ldots, m_i\} \\ &\text{and } \Delta_i = \operatorname{supp}(D \cap M_i) - \max_{D \subset E \subseteq M} \operatorname{supp}(E \cap M_i) \\ &\quad \text{then } i \leq j \implies \Delta_i(D) \geq \Delta_j(D), \quad \forall D \subseteq M,\ i, j \in \overline{1, n}. \end{aligned} \tag{10}$$

ProtoAlgo An algorithm for mining stable descriptions would be:

- Initialise the set of stable descriptions $\mathbb{D}^0_{\texttt{stable}}$,
- Iterate through attributes $m_i,\ i = 1, \ldots, n$:
  - Add m_i to existing stable descriptions: $\mathbb{D}^i_{\texttt{new}} = \{D \cup \{m_i\} \mid D \in \mathbb{D}^{i-1}_{\texttt{stable}}\}$
  - Evaluate Δ_i for each old and new description from $\mathbb{D}^i_{\texttt{cand}} = \mathbb{D}^{i-1}_{\texttt{stable}} \cup \mathbb{D}^i_{\texttt{new}}$
  - Save new stable descriptions $\mathbb{D}^i_{\Delta} = \{D \in \mathbb{D}^i_{\texttt{cand}} \mid \Delta_i(D) \geq \Delta_{\min}\}$

Limitations of Δ-stability

Δ-stability measure $\Delta(D)$ does not depend just on the description, but also on the attributes M surrounding this description: the more attributes are "similar" the fewer there are stable descriptions.

This is especially important for numerical data. For example, take any geographical data with longitude dimension long. Attributes

$$M = \{\text{long} \geq 40, \text{long} \geq 40.01, \text{long} \geq 40.02, \ldots, \text{long} \geq 49.99, \text{long} \geq 50\}$$

will give no stable description because they describe very similar subsets of objects.

So we *have to choose* some discretisation:

$$M = \{\text{long} \geq 40, \text{long} \geq 42, \text{long} \geq 44, \text{long} \geq 46, \text{long} \geq 48, \text{long} \geq 50\}.$$

And the more dimensions there are in the data, the harsher should be the discretisation!

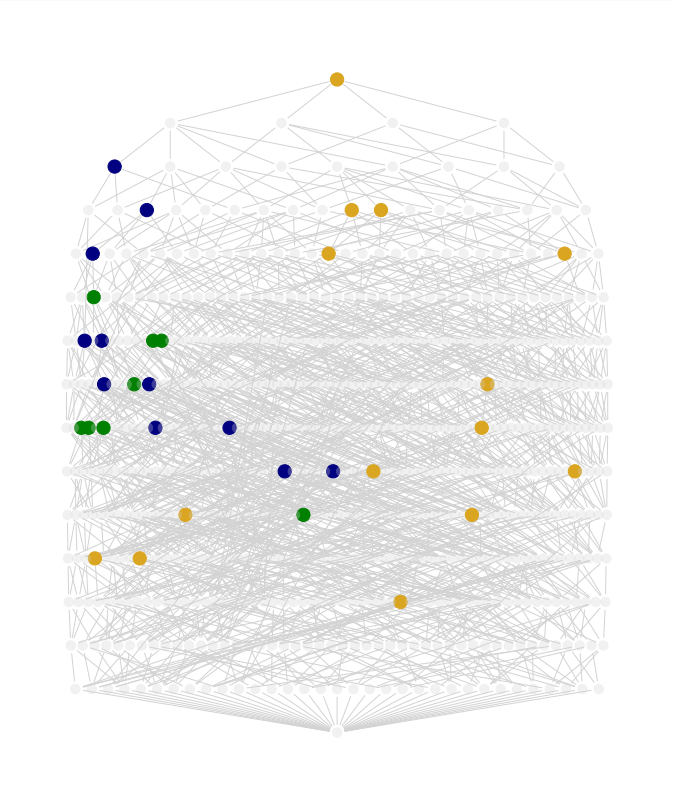
Classifying descriptions

Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

$$|L| \leq 2^{|M|}.$$

Solution Mine only *classifying* concepts and descriptions that help predicting the target value.

Classifying concepts



Classifying descriptions (desc.)

Binary classification Every description D is a binary classifying in itself: it "predicts" positive labels for objects from its extent D' and negative labels for all other objects $G \setminus D'$. So, given two object classes G_+ and $G_- = G \setminus G_+$ a possible interestingness measure could be:

$$\forall D \subseteq M, \mathtt{F1}(D) = \frac{2|D' \cap G_+|}{2|D' \cap G_+| + |D' \setminus G_+| + |G_+ \setminus D'|} \tag{11}$$

Subgroup Discovery Another measure would be Weighted Relative Accuracy score (WRAcc score) popular in Subgroup Discovery community:

$$\mathtt{WRAcc}(D) = \frac{|D'|}{|G|}\Big(\frac{|G_+ \cap D'|}{|D'|} - \frac{|G_+|}{|G|}\Big). \tag{12}$$

The first part of this measure gives better scores to frequent descriptions, while the second part prefers descriptions that predict the target attribute better than the average.

Here comes FCA Both `F1` and `WRAcc` depend only on the extent of description D. So both minimal and maximal descriptions of the same equivalence class will receive the same score.

JSM-method[4] of hypothesis generation & FCA based classification

Consider a **target attribute** $t \notin M$.

- **Positive examples:** Set $G_+ \subseteq G$ of objects known to have t,
- **Negative examples:** Set $G_- \subseteq G$ of objects known not to have t,
- **Undetermined examples:** Set $G_\tau \subseteq G$ of objects for which it is unknown whether they have the target attribute or do not have it.

Consider the three subcontexts of $\mathbb{K} = (G, M, I)$: $\mathbb{K}_\varepsilon := (G_\varepsilon, M, I_\varepsilon)$, where $\varepsilon \in \{-, +, \tau\}$, with respective derivation operators $(\cdot)^+$, $(\cdot)^-$, and $(\cdot)^\tau$.

A **positive hypothesis** $H \subseteq M$ is an intent of $\mathbb{K}_+$ not contained in the intent g^- of any negative example $g \in G_-$: $\forall g \in G_- \quad H \not\subseteq g^-$.

Equivalently: $H^{++} = H, \quad H' \subseteq G_+ \cup G_\tau$.

Also: An intent of $\mathbb{K}_+$ that is contained in the intent of a negative example is called a **falsified (+)-generalisation**

[4] John Stuart Mill, the philosopher who proposed schemes of inductive reasoning (19th century)

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Equivalently: $H^{++} = H, \quad H' \subseteq G_+ \cup G_\tau$.

Also: An intent of $\mathbb{K}_+$ that is contained in the intent of a negative example is called a **falsified (+)-generalisation**

[4] John Stuart Mill, the philosopher who proposed schemes of inductive reasoning (19th century)

Example of a learning context

	G \ M	color	firm	smooth	form	**fruit**
1	apple	yellow	no	yes	round	+
2	grapefruit	yellow	no	no	round	+
3	kiwi	green	no	no	oval	+
4	plum	blue	no	yes	oval	+
5	toy cube	green	yes	yes	cubic	−
6	egg	white	yes	yes	oval	−
7	tennis ball	white	no	no	round	−

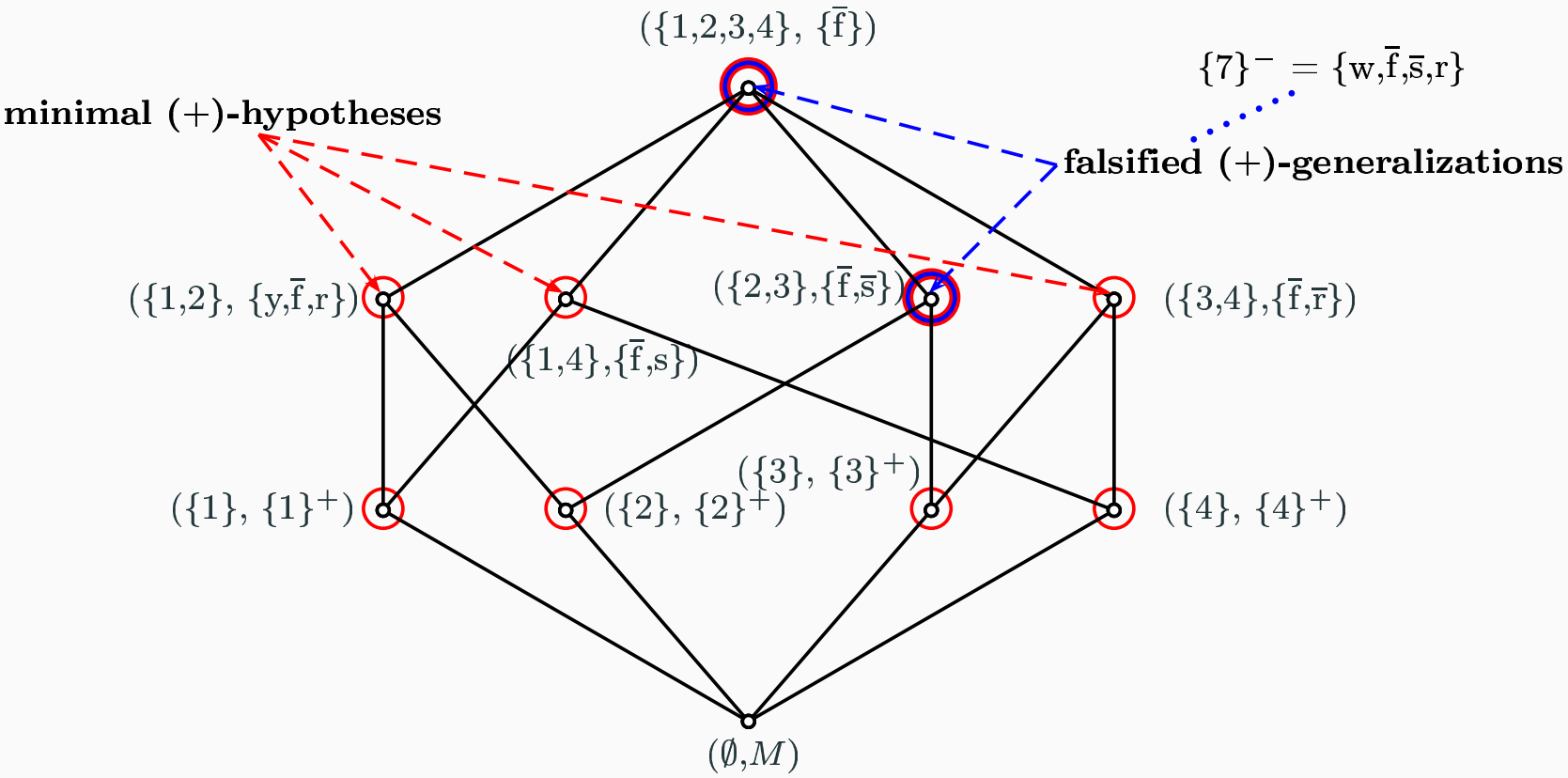
Natural scaling of the context

	G \ M	w	y	g	b	f	$\overline{f}$	s	$\overline{s}$	r	$\overline{r}$	**fruit**
1	apple		×				×	×		×		+
2	grapefruit		×				×		×	×		+
3	kiwi			×			×		×		×	+
4	plum				×		×	×			×	+
5	toy cube			×		×		×			×	−
6	egg	×				×		×			×	−
7	tennis ball	×					×		×	×		−

Abbreviations:

- "g" for green, "y" for yellow, "w" for white, "b" for blue,
- "f" for firm, "$\overline{f}$" for nonfirm, "s" for smooth, "$\overline{s}$" for nonsmooth, "r" for round, "$\overline{r}$" for nonround

Positive Concept Lattice



	G \ M	w	y	g	b	f	$\bar{f}$	s	$\bar{s}$	r	$\bar{r}$	**fruit**
1	apple		×				×	×		×		+
2	grapefruit		×				×		×	×		+
3	kiwi			×			×		×		×	+
4	plum				×		×	×			×	+
5	toy cube			×		×		×			×	−
6	egg	×				×		×			×	−
7	tennis ball	×					×		×	×		−

Hypotheses vs. implications

A positive hypothesis H corresponds to an implication $H \to \{t\}$ in the context $K_+ = (G_+, M \cup \{t\}, I_+ \cup G_+ \times \{t\})$.

A negative hypothesis H corresponds to an implication $H \to \{\bar{t}\}$ in the context $K_- = (G_-, M \cup \{\bar{t}\}, I_- \cup G_- \times \{\bar{t}\})$.

Hypotheses are special implications: their premises are closed (in K_+ or in K_-).

	G \ M	w	y	g	b	f	$\bar{f}$	s	$\bar{s}$	r	$\bar{r}$	**fruit**	**nonfruit**
1	apple		×				×	×		×		×	
2	grapefruit		×				×		×	×		×	
3	kiwi			×			×		×		×	×	
4	plum				×		×	×			×	×	
5	toy cube			×		×		×			×		×
6	egg	×				×		×			×		×
7	tennis ball	×					×		×	×			×

Classification of an undetermined example g_τ

g_τ^τ is **classified positively** if it contains a positive hypothesis and does not contain any negative hypothesis (the presence of the goal property t is predicted)

g_τ^τ is **classified negatively** if it contains a negative hypothesis and does not contain any positive hypothesis (the absence of the goal property t is predicted)

g_τ^τ is **contradictory** if it contains both positive and negative hypotheses, then it is said that the classification is *contradictory*.

NB: If g_τ^τ does not contain any negative or positive hypotheses as subsets, the classification does not take place.

Classification of an undetermined example g_τ

	G \ M	color	firm	smooth	form	**fruit**
1	apple	yellow	no	yes	round	$+$
2	grapefruit	yellow	no	no	round	$+$
3	kiwi	green	no	no	oval	$+$
4	plum	blue	no	yes	oval	$+$
5	toy cube	green	yes	yes	cubic	$-$
6	egg	white	yes	yes	oval	$-$
7	tennis ball	white	no	no	round	$-$
8	mango	yellow	no	yes	oval	τ

Classification of an undetermined example mango

	G \ M	w	y	g	b	f	$\overline{f}$	s	$\overline{s}$	r	$\overline{r}$	**fruit**
1	apple		×				×	×		×		+
2	grapefruit		×				×		×	×		+
3	kiwi			×			×		×		×	+
4	plum				×		×	×			×	+
5	toy cube			×		×		×			×	−
6	egg	×				×		×			×	−
7	tennis ball	×					×		×	×		−
8	mango		×				×	×			×	τ

Mango is classified positively:

- for (+)-hypothesis $\{\overline{r}, \overline{f}\}$: $\{\overline{r}, \overline{f}\} \subseteq mango^{\tau} = \{y, \overline{f}, s, \overline{r}\}$;
- for (−)-hypothesis $\{w\}$ and $\{f, s, \overline{r}\}$: $\{w\} \nsubseteq mango^{\tau}$ & $\{f, s, \overline{r}\} \nsubseteq mango^{\tau}$.

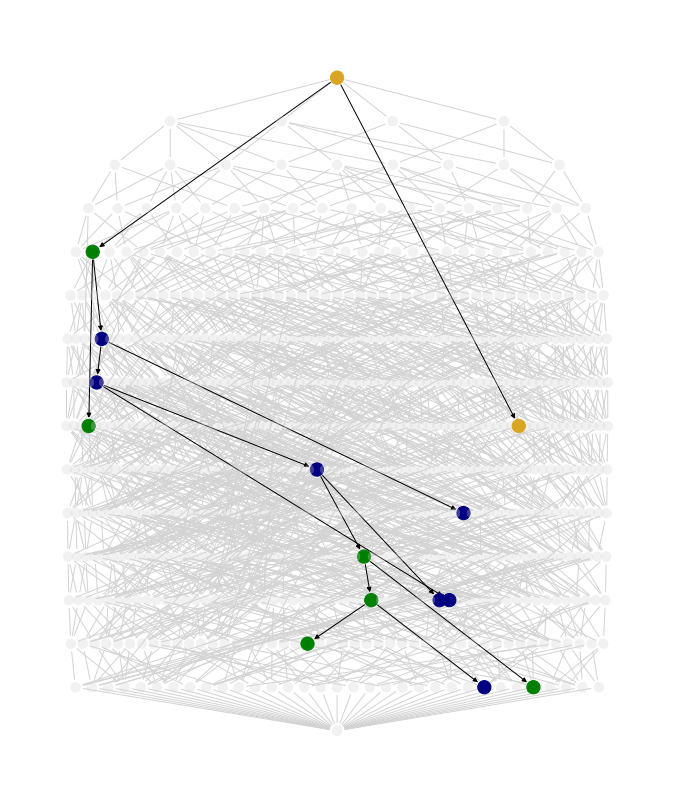
Decision trees

Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

$$|L| \leq 2^{|M|}.$$

Solution Mine *decision trees*, i.e. meet-semilattices of minimal subsets of attributes.

Decision tree



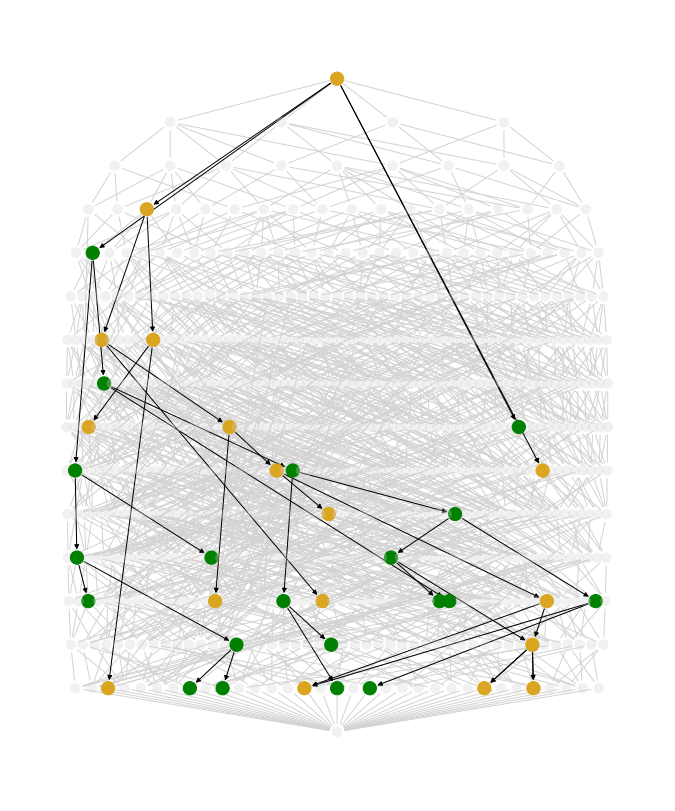
Ensembles of decision trees

Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

$$|L| \leq 2^{|M|}.$$

Solution Mine *ensembles of decision trees*, i.e. multiple meet-semilattices of minimal subsets of attributes.

Random Forest



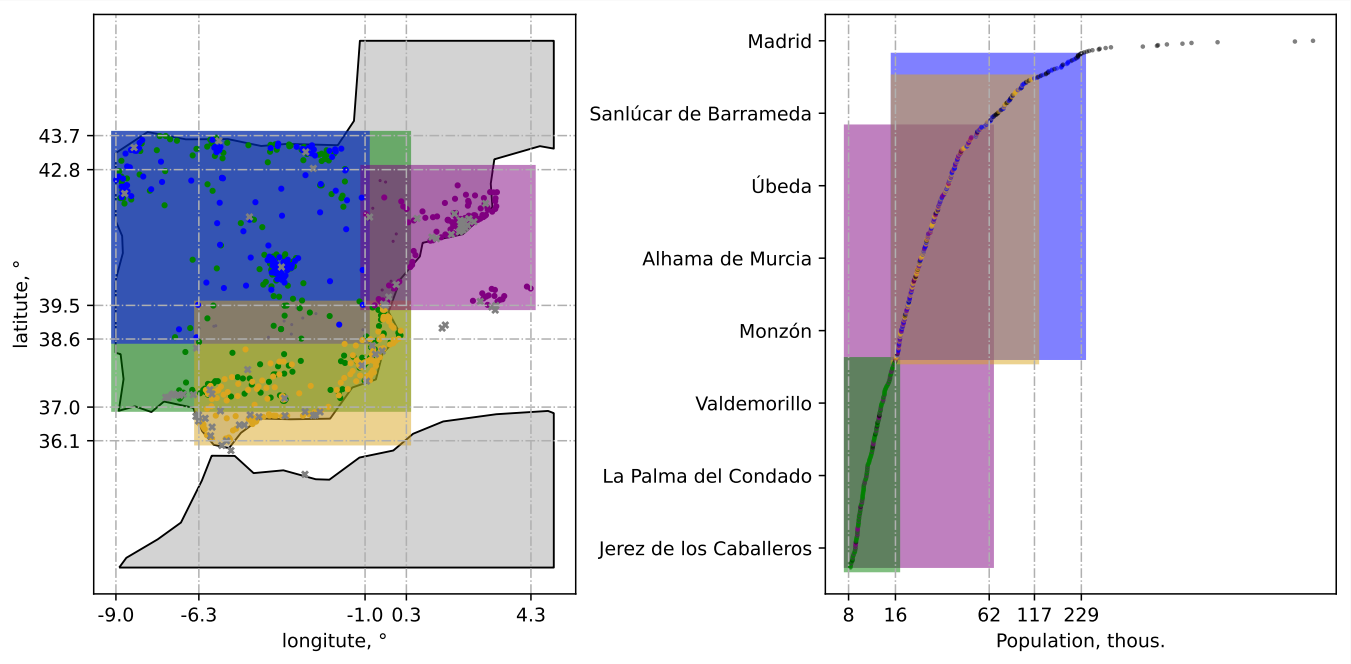
Clusterings

Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

$$|L| \leq 2^{|M|}.$$

Solution Mine *clusterings*, i.e. few subsets of attributes, that do not overlap and cover almost all the data.

Clustering of cities of Spain



Clusterings (cont.)

Clustering Say that every attribute M is a cluster. Then a *perfect clustering* is a subset of attributes C whose extents, together, cover all the data while not overlapping:

$$C \subseteq M \text{ is a perfect clustering} \iff \bigcup_{m \in C} m' = G, \text{ and } m_1' \cap m_2' = \emptyset, \ \forall m_1, m_2 \in M \tag{13}$$

Perfect clusterings are hard to obtain. So we search for more relaxed solutions.

Broad non-overlapping clustering A clustering $C \subseteq M$ is called θ-*broad* if it covers more θ objects. A clustering is called η-*non-overlapping* if the pairwise overlap between each of its clusters is less than η:

$$\begin{aligned} \text{Clustering } C \text{ is } \theta\text{-broad} &\iff \left| \bigcup_{m \in C} m' \right| \geq \theta. \\ \text{is } \eta\text{-non-overlapping} &\iff |m_1' \cap m_2'| \leq \eta, \forall m_1, m_2 \in M. \end{aligned} \tag{14}$$

Here comes FCA A clustering as a minimal broad clustering if and only if the set of its "residuals" is a minimal rare description in the context of residuals!

Algorithm for mining minimal rare descriptions

SubClustering property Every subset of a minimal non-overlapping clustering is a non-overlapping clustering:

$$\forall C,\ D \text{ is a non-overlapping clustering} \implies E \text{ is a non-overlapping clustering}, \forall E \subset C \tag{15}$$

ProtoAlgo So, the generic algorithm for mining broad non-overlapping clusterings would be:

- Put the empty clustering into a queue $Q = \langle \emptyset \rangle$
- While the queue is not empty ($|Q| > 0$):
  - Take clustering from the queue $C = Q.\mathrm{pop}()$
  - Test if C is a minimal non-overlapping clustering, i.e. every smaller $E \subset C$ covers fewer objects and has no overlapping,
  - If C is minimal non-overlapping, test if it is broad,
  - If D is minimal non-overlapping and not broad, add next clusterings to the queue $\forall m \in M \setminus C,\ Q.\mathrm{add}(C \cup \{m\})$

The algorithm is inspired by the algorithm for mining minimal rare descriptions.

Conclusion

Problem Given formal context (G, M, I) with many objects G and fewer attributes M, there can be as many concepts as there are subsets of attributes:

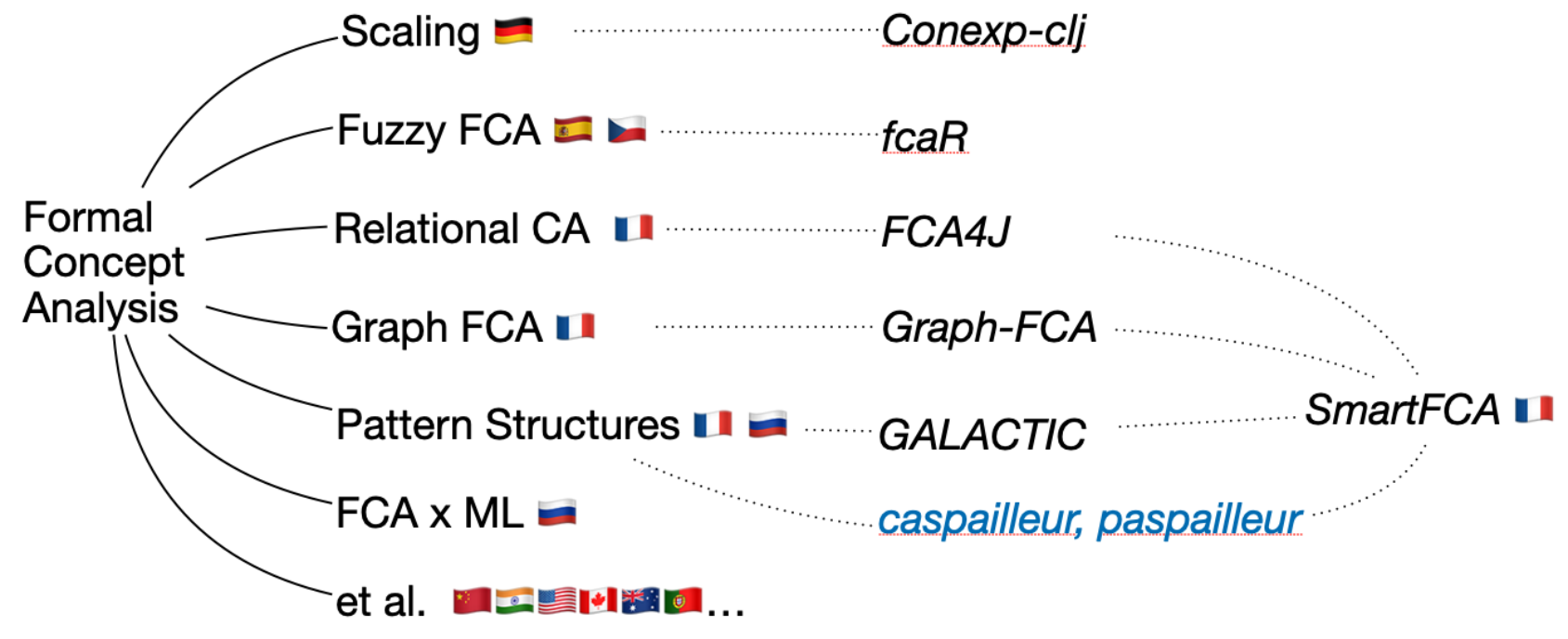
$$|L| \leq 2^{|M|}.$$

Solution Mine only *interesting* concepts and descriptions. Such as:

- Maximal frequent descriptions,
- Minimal rare descriptions,
- Δ-stable descriptions,
- Subgroups,
- (Leaves of) decision trees and ensembles of decision trees,
- Minimal broad non-overlapping clusterings,
- and many others: jumping emerging patterns, etc...

Pattern Structures

Extensions of FCA



Formal Concept Analysis – in its binary form – was actively developing during 1980-2000 years. In 2000s there emerged many different extensions of FCA that specified for various complex data.

NB The diagram is not complete (and can never be), but it gives some overview of the situation.

Three branches of SmartFCA

Pattern Structures

First proposed in 2001: B. Ganter and S. O. Kuznetsov, Pattern Structures and their Projections, ICCS.

Analyses "tabular" data, even there is just one complex columns.

Mostly studied in Nancy and La Rochelle.

Relational Concept Analysis

First proposed in 2007: M. Rouane, M. Huchard, A. Napoli, and P. Valtchev, Proposal for Combining Formal Concept Analysis and Description Logics for Mining Relational Data, ICFCA.

Analyses "relational" data, that are SQL-like databases with multiple related tables.

Mostly studied in Montpellier and Strasbourg.

Graph FCA

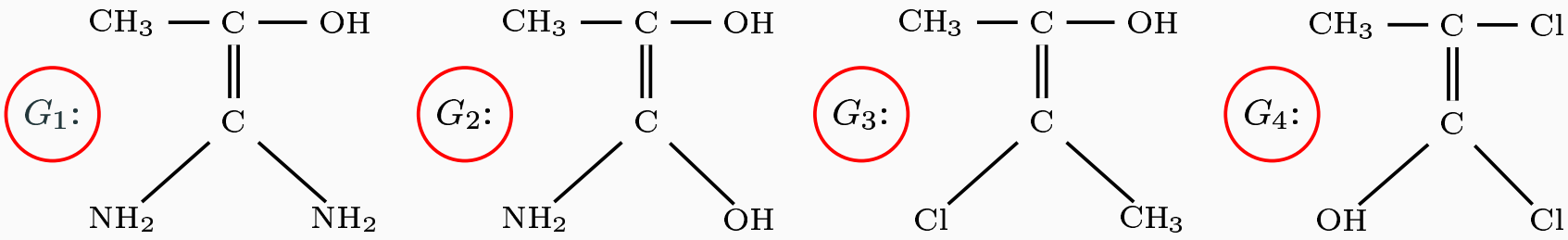
First proposed in 2015: S. Ferré, Ferré, S.: A proposal for extending formal concept analysis to knowledge graphs, ICFCA.

Analyses "graph" data, and search for tuples of objects (i.e. of nodes) with common neighbourhood.

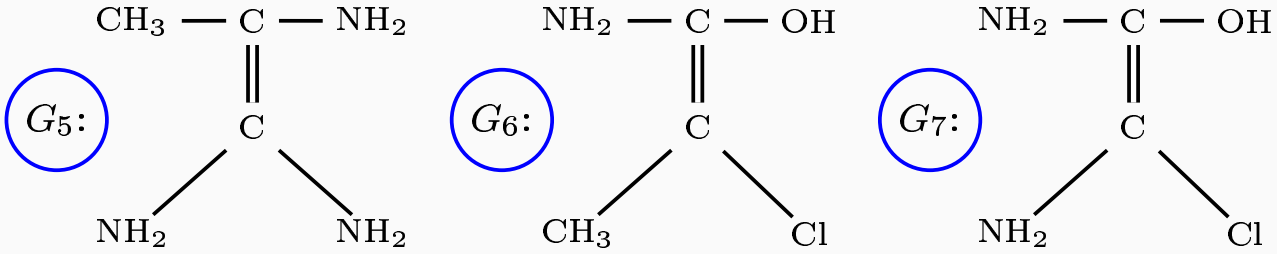
Mostly studied in Rennes and Strasbourg.

Quick Pattern Structure example

Positive examples:

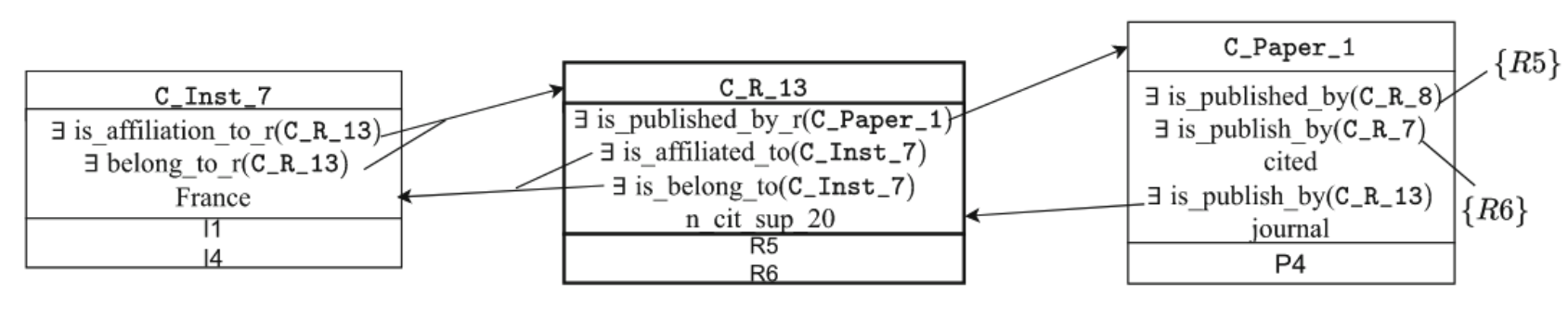


Negative examples:



Quick RCA example

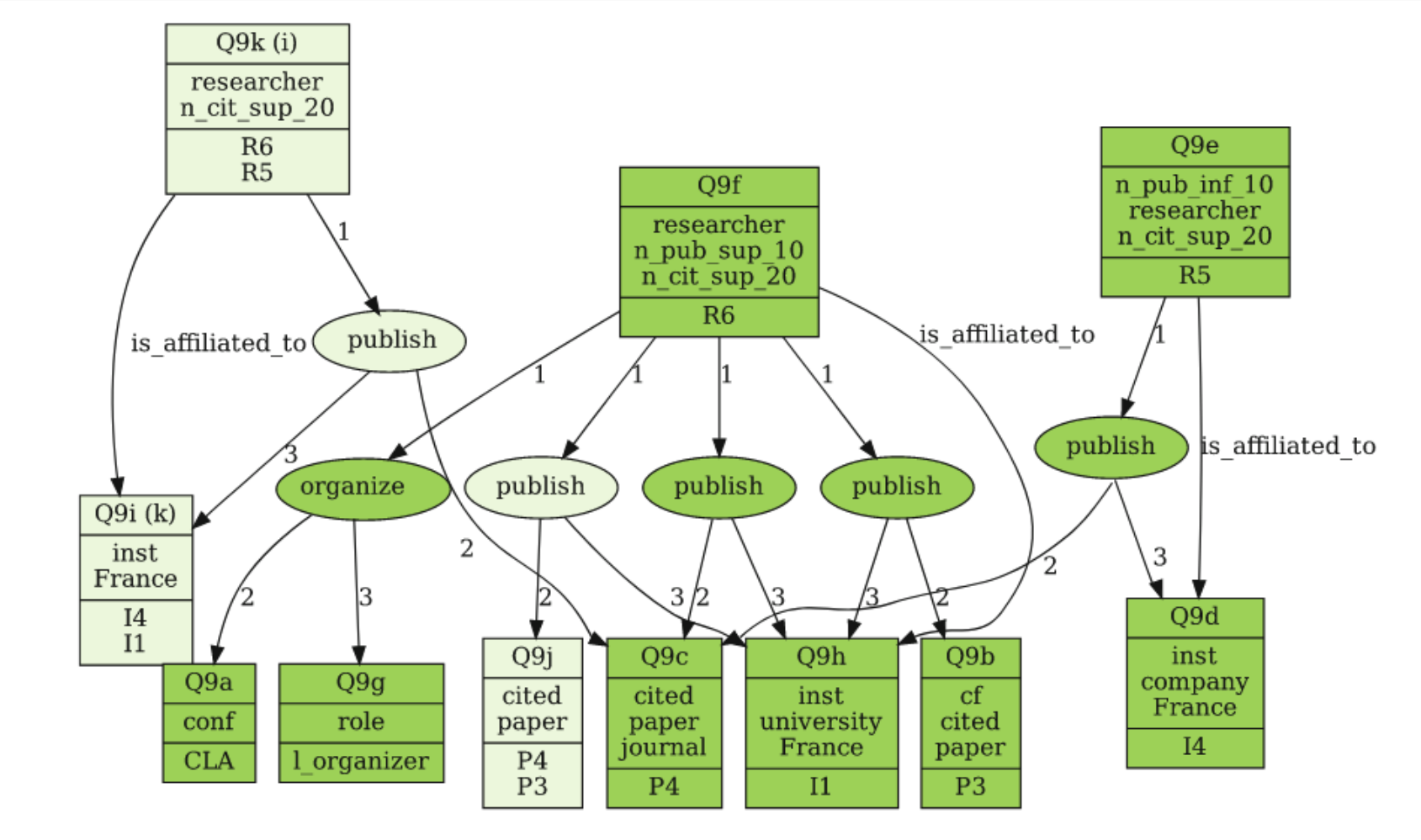
Figure Example of an RCA intent:
Given three datatables on Institutes, Researchers and Papers



Copied from V. Fokou et al. Relational Concept Analysis and Graph-FCA in Practice: Analyzing Multi-relational Data, CONCEPTS 2025.

Quick GraphFCA example

Figure Example of a graph pattern



Copied from V. Fokou et al. Relational Concept Analysis and Graph-FCA in Practice: Analyzing Multi-relational Data, CONCEPTS 2025.

Motivation for Pattern Structures

What if we keep studying tabular data and keep objects "independent".

But allow every object to be described by some complex description?

Recall that in a formal context (G, M, I) we have a lattice of descriptions $\mathbb{D}$:

$$(G, M, I) \iff \mathbb{D} = 2^M. \tag{16}$$

What if we keep the lattice of descriptions $\mathbb{D}$ but "detach" it from binary attributes?

$$(G, ?, ?) \iff \mathbb{D} = ? \tag{17}$$

(Classical) definition of a pattern structure

A pattern structure $(G, (D, \sqcap), \delta)$ is composed of:

- G a set of *objects*,
- $(D, \sqcap)$ a semi-lattice of descriptions or patterns,
- δ a mapping such as $\delta(g) \in D$ describes object g.

The Galois connection for $(G, (D, \sqcap), \delta)$ is defined as:

- The maximal description representing the similarity of a set of objects:

$$A^{\diamond} = \sqcap_{g \in A} \delta(g) \quad \text{for } A \subseteq G$$

- The maximal set of objects sharing a given description:

$$d^{\diamond} = \{g \in G | d \sqsubseteq \delta(g)\} \quad \text{for } d \in (D, \sqcap)$$

Standard FCA as a Pattern Structure $(G, (D, \sqcap), \delta)$

Considering a standard formal context (G, M, I):

- G is the set of *objects*,
- $(D, \sqcap)$ corresponds to $\mathcal{P}(M)$ where M is the set of attributes.
- $\delta(g)$ corresponds to the description of g in terms of attributes.

	m_1	m_2	m_3
g_1	×		×
g_2	×	×	
g_3		×	×
g_4		×	×
g_5	×	×	×

The Galois connection:

- $A^{\diamond} = \sqcap_{g \in A} \delta(g)$ for $A \subseteq G$
- $\{g_1, g_2\}' = g_1' \cap g_2' = \{m_1, m_2\} \cap \{m_1, m_3\} = \{m_1\}$
- $d^{\diamond} = \{g \in G | d \sqsubseteq \delta(g)\}$ for $d \in (D, \sqcap)$
- $\{m_1\}' = \{g_i \in G | \{m_1\} \subseteq g_i'\} = \{g_1, g_2, g_5\}$

From FCA to Pattern Structures

A formal context (G, M, I) is based on a set of objects G, a set of attributes M, and a binary relation $I \subseteq G \times M$.

Two derivation operators defined by:
$\forall A \subseteq G, B \subseteq M$,

$$\begin{aligned} A' &= \{m \in M \mid \forall g \in A, (g, m) \in I\} \\ B' &= \{g \in G \mid \forall m \in B, (g, m) \in I\} \end{aligned}$$

A **formal concept** (A, B) verifies

$$A' = B \quad \text{and} \quad A = B'$$

Formal concepts are partially ordered by inclusion of extents (or intents):

$$(A_1, B_1) \leq (A_2, B_2) \text{ iff } A_1 \subseteq A_2$$

A pattern structure $(G, (\mathcal{D}, \sqcap), \delta)$ is based on G, a meet-semilattice of *descriptions* $(\mathcal{D}, \sqcap)$, and $\delta : G \to \mathcal{D}$ that sends each object to a description.

Two derivation operators are defined by:
$\forall A \subseteq G, d \in (\mathcal{D}, \sqcap)$,

$$\begin{aligned} A^\diamond &= \sqcap_{g \in A} \delta(g) \\ d^\diamond &= \{g \in G \mid d \sqsubseteq \delta(g)\} \end{aligned}$$

A **pattern concept** (A, d) verifies

$$A^\diamond = d \quad \text{and} \quad A = d^\diamond$$

Pattern concepts are partially ordered by inclusion of extents (or intents):

$$(A_1, d_1) \leq (A_2, d_2) \text{ iff } A_1 \subseteq A_2$$

Implicative dependencies in pattern structures

Consider a pattern structure $(G, (\mathcal{D}, \sqcap), \delta)$.

Pattern implication: $d_1 \to d_2$ holds **iff** $d_1^\diamond \subseteq d_2^\diamond$.

Pattern association rule $d_1 \to_{c,s} d_2$ has

confidence $c = \frac{|d_1^\diamond \cap d_2^\diamond|}{|d_1^\diamond|}$ and

support $s = \frac{|d_1^\diamond \cap d_2^\diamond|}{|G|}$

Many types of patterns

Pattern Name	When to use	Example
ItemSet Pattern	tags and keywords	{CNN, DL} ⊓ {GNN, DL} = {DL}
CategorySet Pattern	categorical data	{by MosAlk} ⊓ {by Resist} = {by MosAlk, by Resist}
SequenceSet Pattern	sequences	{“Deep Reinforcement Learning”} ⊓ {“Deep Learning”} = {“Deep”, “Learning”}
Interval Pattern	numerical data	#Words: [9, 10] ⊓ #Words: [11, 15] = #Words: [9, 15]
Cartesian Pattern	tabular data	({CNN, DL}, [9, 10]) ⊓ ({GNN, DL}, [11, 15]) = ({DL}, [9, 15])
GraphSet Pattern	graphs	{graph X} ⊓ {graph Y} = maximal common connected induced subgraphs of X and Y

Types of Patterns available in Paspailleur software v0.1.

Graph pattern structures

Idea: take graphs as objects (described by sets of attributes as for contexts)

Thus: we need to define a *similarity* $\sqcap$ on graphs, similar to set-theoretic $\cap$

First natural attempt: take the largest common subgraph of two graphs

However: it fails since there can be several of such subgraphs

Second natural attempt: define similarity for sets of graphs

However: it also fails even if we take all largest (in the number of vertices) common subgraphs of two graphs

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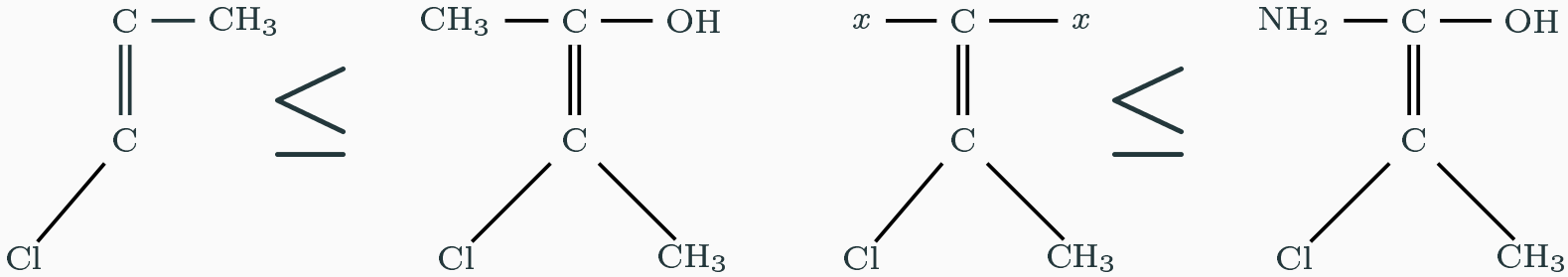
Order on labeled graphs

Let $(\mathcal{L}, \preceq)$ be an ordered set of vertex labels.

Definition: $\Gamma_1 := ((V_1, l_1), E_1)$ **is dominated by** $\Gamma_2 := ((V_2, l_2), E_2)$, $\Gamma_1 \leq \Gamma_2$, if there exists a graph *homomorphism* $\varphi\colon V_1 \to V_2$, i.e.,

- preserves edges: $(v, w) \in E_1 \Rightarrow (\varphi(v), \varphi(w)) \in E_2$,
- fits under labels: $l_1(v) \preceq l_2(\varphi(v)$.

Example: $\mathcal{L} = \{x, NH_2, Cl, CH_3, C, OH\}$ with $x \preceq A \in \mathcal{L}$ and $A||B$ if $A, B \neq x$.



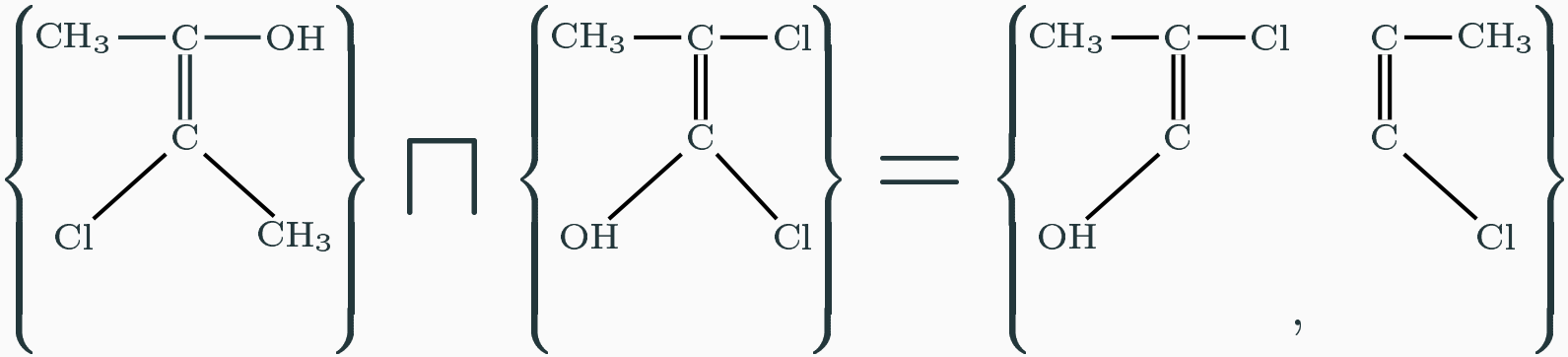
vertex labels are unordered

$x \preceq A$ for any vertex label $A \in \mathcal{L}$

Similarity:

$$\begin{aligned} \{X\} \sqcap \{Y\} &:= \{Z \mid Z \leq X, Y, \quad \forall Z_* \leq X, Y,\ Z_* \not\leq\ Z\} \\ &= \text{set of all maximal common subgraphs of } X \text{ and } Y \end{aligned}$$

Example:



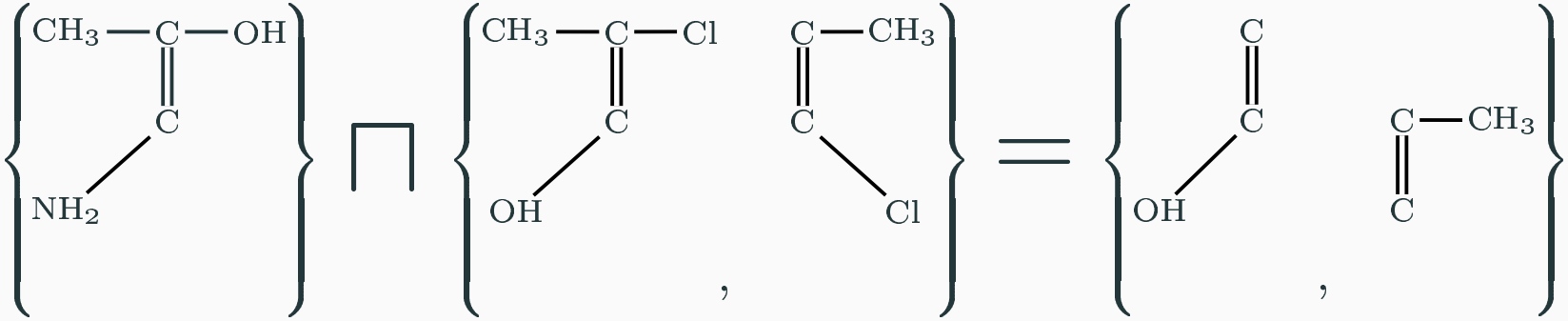
Meet of graph sets

Extension: For sets of graphs $\mathcal{X} = \{X_1, \ldots, X_k\}$ and $\mathcal{Y} = \{Y_1, \ldots, Y_n\}$, define:

$$\mathcal{X} \sqcap \mathcal{Y} := \ \mathrm{MAX}_{\leq}\Big\{ \bigcup_{i,j} \big(\{X_i\} \sqcap \{Y_j\} \big) \Big\}$$

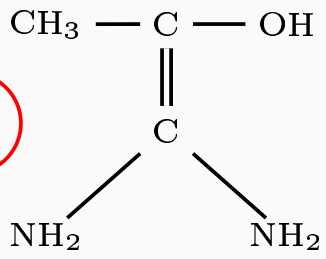
NB: $\sqcap$ is idempotent, commutative, and associative.

Example:

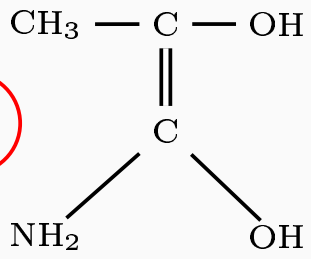


Positive examples:

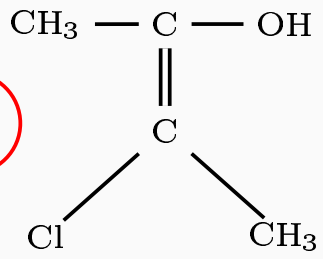
G_1:



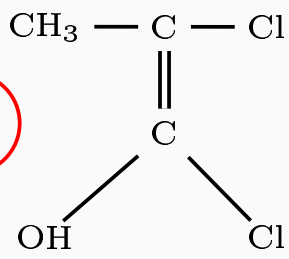
G_2:



G_3:

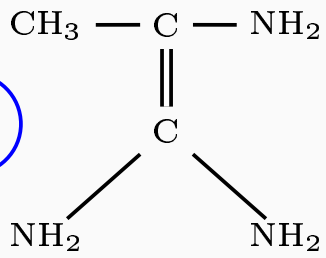


G_4:

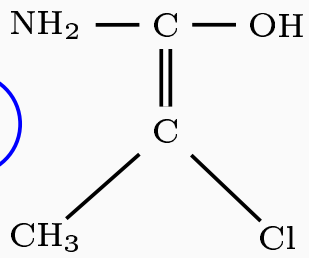


Negative examples:

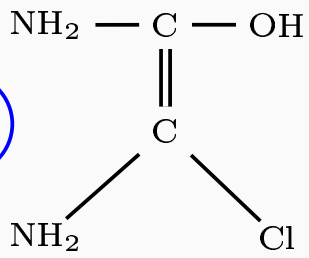
G_5:



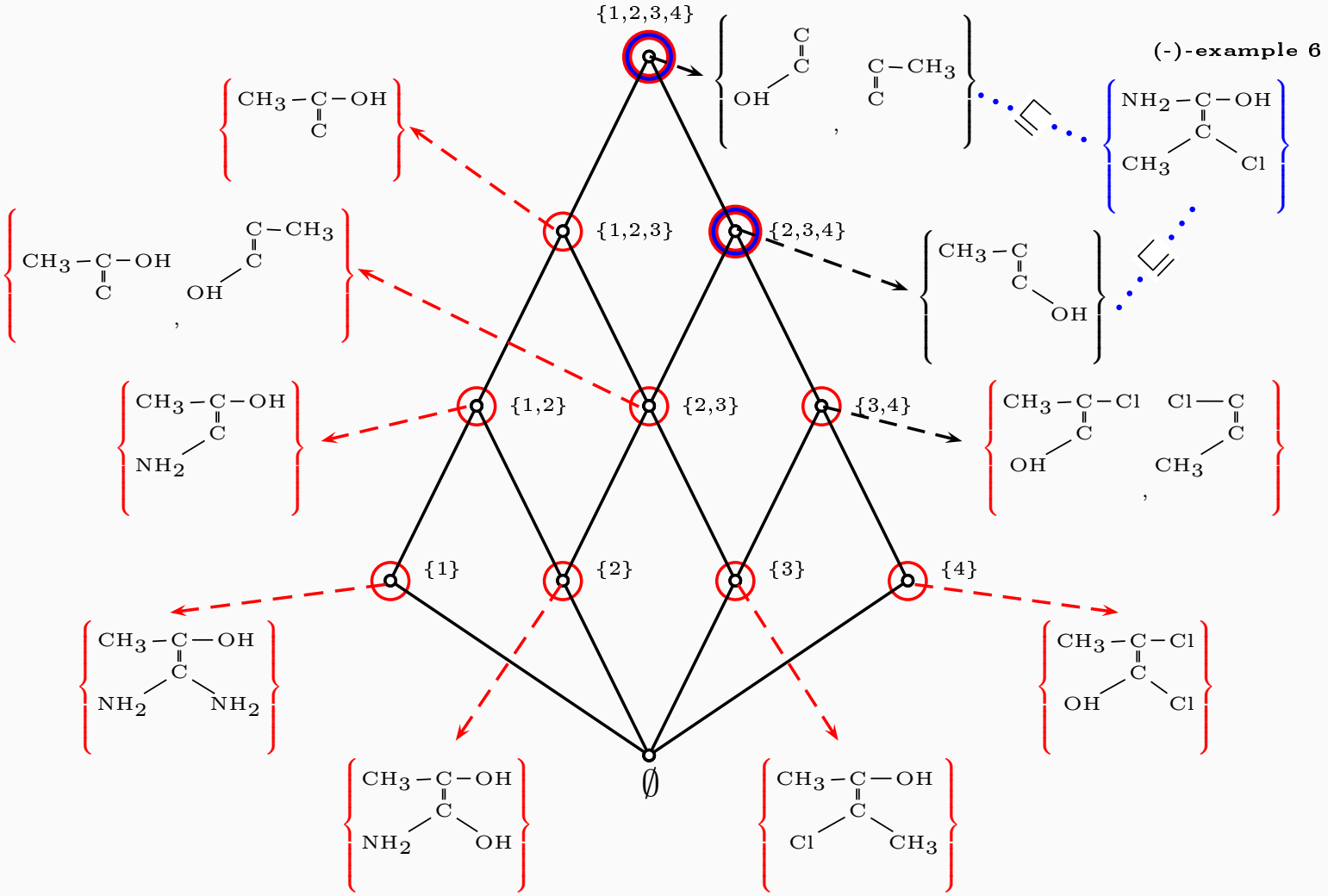
G_6:



G_7:



Positive lattice



positive examples 1, 2, 3, 4

Further applications of graph structures

- Predicting biological properties of chemical compounds
- Designing chemical compounds with desired properties
- Analysis of chemical reactions

- Text/argument clustering and classification
- Code verification and alignment (par ex. A.S.T.)
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NLP Application of Graph Patterns: Clustering Reviews

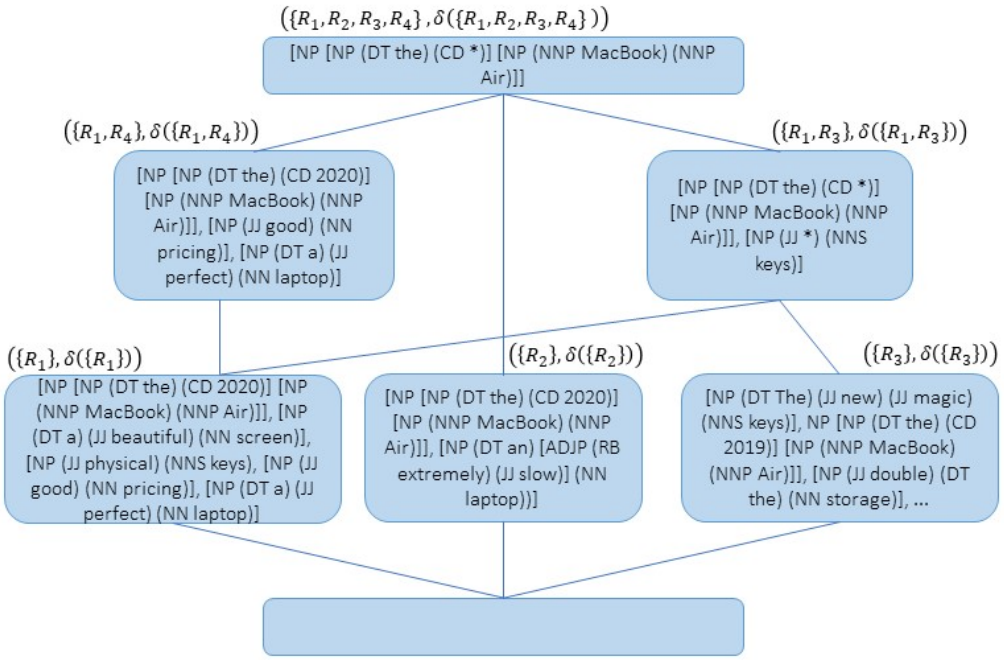
R_1: *In my opinion, **the 2020 MacBook Air** is **a perfect laptop**. It has **physical keys** that are a pleasure to type on, **a beautiful screen, an optimal size, good pricing**, and **a quad-core processor**.*

R_2: *The **MacBook Air** is **an extremely slow laptop**. Right out the box it was slow and even after updating it is still slow and seems to only be getting slower day by day.*

R_3: *The **new magic keys** are amazing when typing. Unlike **the 2019 MacBook Air**. It's so amazing how you pay $999 for **double the storage** you got form last year that's starts at $1099. It is very fast and smooth.*

R_4: *The **2020 MacBook Air** is **a perfect laptop** due to its **good pricing**.*

Diagram of the pattern lattice for the reviews

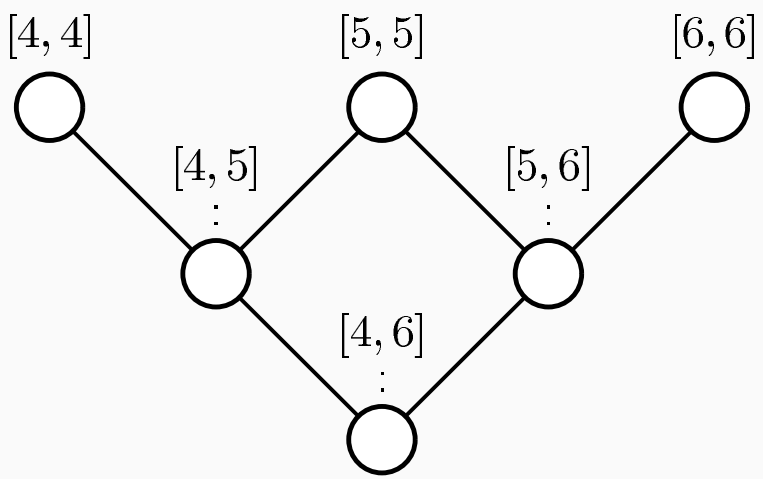


Example: Interval Pattern Structure

- Let $\mathcal{D}$ be a set of intervals with integer bounds (for simplicity)
- Let $\sqcap$ be a meet operator defined on $\mathcal{D}$ as the convex hull of intervals:

$$\begin{aligned} [a_1, b_1] \sqcap [a_2, b_2] &= [min(a_1, a_2), max(b_1, b_2)] \\ [4, 5] \sqcap [5, 5] &= [4, 5] \end{aligned}$$

$$\begin{aligned} [a_1, b_1] \sqsubseteq [a_2, b_2] &\iff [a_2, b_2] \subseteq [a_1, b_1] \\ [4, 5] \sqsubseteq [5, 5] &\iff [5, 5] \subseteq [4, 5] \end{aligned}$$



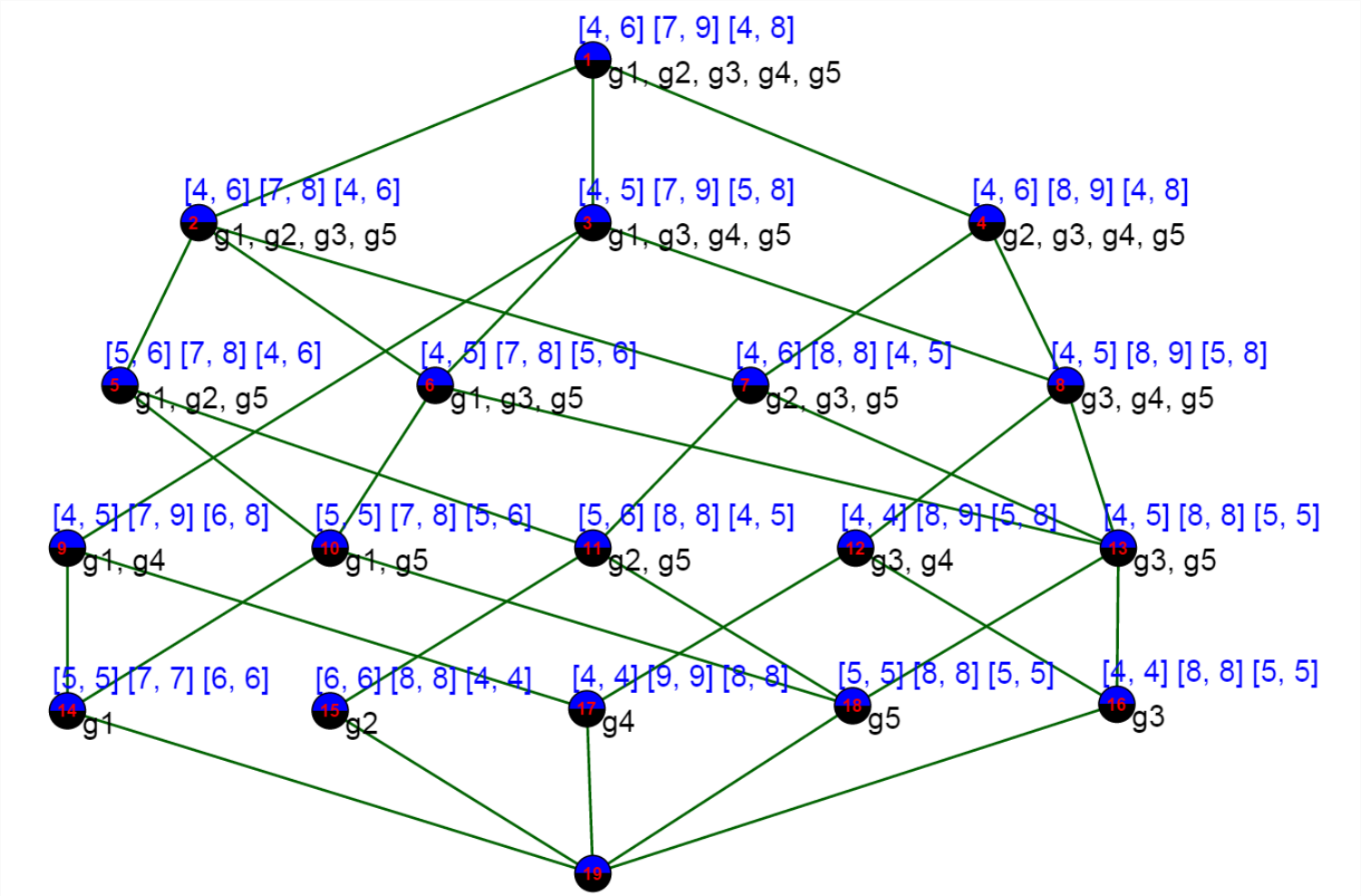
Interval Pattern Structures for classifying a numerical context

	m_1	m_2	m_3
g_1	5	7	6
g_2	6	8	4
g_3	4	8	5
g_4	4	9	8
g_5	5	8	5

$$\begin{aligned}
\{g_1, g_2\}^\diamond &= \sqcap_{g \in \{g_1, g_2\}} \delta(g) \\
&= \langle 5, 7, 6 \rangle \sqcap \langle 6, 8, 4 \rangle \\
&= \langle [5, 6], [7, 8], [4, 6] \rangle \\
\langle [5, 6], [7, 8], [4, 6] \rangle^\diamond &= \{g \in G | \langle [5, 6], [7, 8], [4, 6] \rangle \sqsubseteq \delta(g)\} \\
&= \{g_1, g_2, g_5\}
\end{aligned}$$

Example: $(\{g_1, g_2, g_5\}, \langle [5, 6], [7, 8], [4, 6] \rangle)$ is a pattern concept

Interval pattern concept lattice



Highest concepts: largest extents and smallest intents (but the largest intervals)

Lowest concepts: smallest extents and largest intents (but the smallest intervals)

Problem: efficient pattern mining!

Problems of the classical definitions

There are some problems with the "classical" definition of Pattern Structures:

1. When creating new pattern structure $(G, (\mathbb{D}, \sqcap), \delta)$, it is not always obvious if some proposed operation $\sqcap$ is indeed an infimum operation.
2. The meet operation $\sqcap$ supposes that we find concepts in "bottom-up" approach: we start with concepts on 1 object, then compute concepts with 2, 3, etc objects. Therefore, to arrive to a frequent concept we should pass by many non-frequent concepts.
3. Some "top-down" pattern structure formulations were proposed for specific data: graphs, intervals, sequences. But how to generalise it to any complex data?

(Novel) definition of pattern structures

Before A pattern structure $(G, (D, \sqcap), \delta)$ is composed of:

- G a set of *objects*,
- $(D, \sqcap)$ a (complete meet) semi-lattice of descriptions or patterns,
- δ a mapping such as $\delta(g) \in D$ describes object g.

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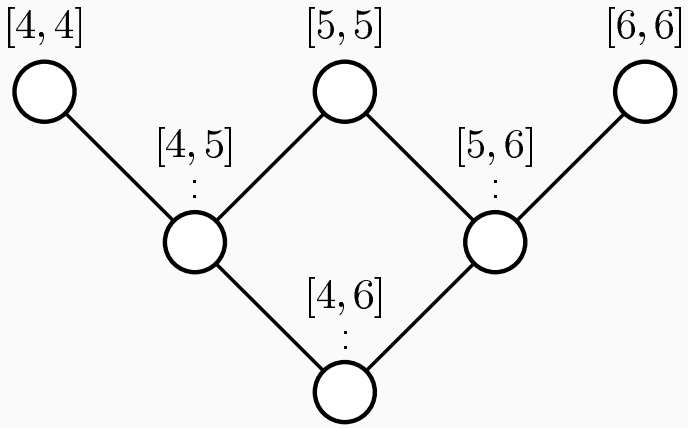
Revelation If $(D, \sqcap)$ is a lattice, then it corresponds to a system of implications Σ. For example, to a system of "apriori" (or "obvious") implications on attributes:

$$(\mathbb{D}, \sqcap) \text{ is a pattern lattice } \iff (M, \Sigma_M) \text{ is an implication system on attributes} \tag{18}$$

A (semi-)lattice of patterns



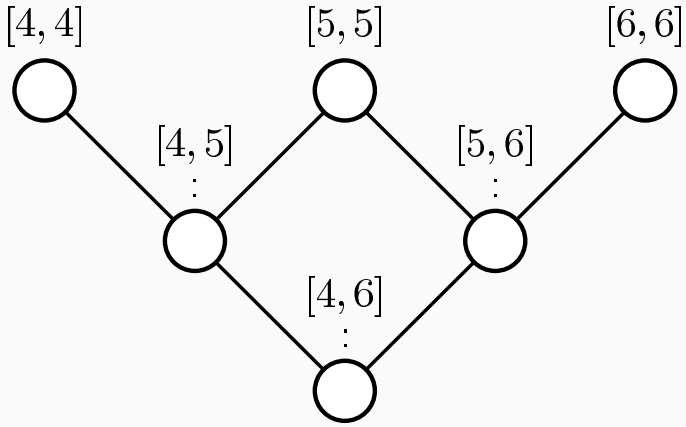
NB The maximal pattern – the emptyset $\emptyset$ that works as a controversy $\perp$ – in not shown on the figure, but it is assumed to be there.

A system of implications

$$M = \{[4,6],[4,5],[5,6],[4,4],[5,5],[6,6],\perp\}$$

$$\Sigma_M = \begin{cases} [4,4] \implies [4,5],[4,6] \\ [5,5] \implies [4,5],[5,6],[4,6] \\ [6,6] \implies [5,6],[4,6] \\ [4,5] \implies [4,6] \\ [5,6] \implies [4,6] \\ [4,5],[5,6] \implies [5,5] \\ [4,4],[5,5] \implies \perp \\ [5,5],[6,6] \implies \perp \end{cases} \tag{19}$$

A (semi-)lattice of patterns



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Highlighted are atomic patterns: all other patterns can be derived from subsets of atomic patterns.

A system of implications

$$M = \begin{cases} [4,6], [4,5], [5,6], [4,4], [5,5], [6,6], \perp, \\ \geq 4, \geq 5, \geq 6, \leq 4, \leq 5, \leq 6 \end{cases}$$

$$\Sigma_M = \begin{cases} \geq 6 \implies \geq 5, \geq 4 \\ \geq 5 \implies \geq 4 \\ \leq 4 \implies \leq 5, \leq 6 \\ \leq 5 \implies \leq 6 \\ \geq 4, \leq 6 \implies [4,6], \\ \geq 4, \leq 5 \implies [4,5], \\ \geq 4, \leq 4 \implies [4,4], \\ \geq 5, \leq 6 \implies [5,6], \\ \geq 5, \leq 5 \implies [5,5], \\ \geq 5, \leq 4 \implies \perp, \\ \geq 6, \leq 6 \implies [6,6], \\ \geq 6, \leq 5 \implies \perp, \\ \emptyset \implies \geq 4, \leq 6, \\ [4,4] \implies \geq 4, \leq 4 \\ [5,5] \implies \geq 5, \leq 5 \\ [6,6] \implies \geq 6, \leq 6 \\ [4,5] \implies \geq 4, \leq 5 \\ [5,6] \implies \geq 5, \leq 6 \\ [4,6] \implies \geq 4, \leq 6 \end{cases} \tag{20}$$

Many new patterns

Now, in order to create a new pattern structure one has to create new pattern lattice $(\mathbb{D}, \sqsubseteq)$.

There are many ways to define a lattice of patterns:

- Provide binary attributes M and implications on them Σ_M,
- Provide binary attributes M and *mine* implications on them Σ_M,
- Provide objects descriptions $\delta(g), \forall g \in G$ and a meet operation $\sqcap$ on them,
- Provide atomic patterns $\mathbb{A}$ and a join operation $\sqcup$ on them,
- Provide all patterns $\mathbb{D}$ and their lattice order $\sqsubseteq$ explicitly.

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Back to reality

I have started implementing the new view on Pattern Structures in Paspailleur v0.2. But the work is not yet finished.

So, for now, on February 2026, it is better to use the last-years version v0.1 of the package: it is not as versatile, but it works fast. And it supports numerical, categorical, and ngrams data.
`https://github.com/smartFCA/paspailleur`

And there is always Caspailleur for binary data:
`https://github.com/smartFCA/caspailleur`.